The DICTO Concordance

A map and almanac of the work: threads, names, concepts and the one triangle

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The work, its ideas and its names are DICTO's, as each paper's credit line says; this map only gathers them.

> *This is a map of DICTO's work in one place, so that no detail has to be carried in the head.* ***Threads*** *follow one idea from paper to paper;* ***Names*** *lists every named solid, piece, family and rule with who named it and where it first appears; the* ***Almanac of Concepts*** *says what each operation, law and method is and where it is used; the last page puts every thread on the one triangle of mirrors. To find anything, look it up in Names or the Almanac and follow the paper numbers.*

1. Threads

11 threads run through the papers. Each paper is listed, in the order of its number, with one line on what it adds to the thread; a paper can sit on several threads. Lines marked → are the verified cross-links of the audit of 4 November 2026 (study 2026-11-04-cross-links, section A).

1.1 Kepler, Euclid and the stars

Euclid's dodecahedron as a cube with six roofs, all five Platonic solids in one cubic cell, and the jewels and stars that fill its gaps.

- **#8** The Euclid–Kepler–Pacioli Cell: all five Platonic solids nest on 13 points in one periodic cell: Euclid's roofs, reflected back through the cube, land on an icosahedron.
- **#10** DICTO Jewel (EKP windows): finding: the DICTO Jewel, with 12 Penrose rhombus windows; with stella octangulas it fills space.
- **#12** Star Chain Reaction: finding: great star, Sunstar, great star, Sunstar, nested exactly, each step touching.
- **#13** The DICTO Dodeca-13: the 13-dodecahedron cluster made solid: its gaps closed by 30 wedges and 20 needles.
- **#14** The DICTO-Star: an icosidodecahedron, 12 dodecahedra and 20 tridiminished icosahedra closing exactly at every edge.
- **#15** DICTO Jewel clusters: finding: four and six DICTO Jewels as clusters, each filling space with stella octangulas.
- **#16** Sunstar clusters: finding: six dodecahedra round a Dogstar, a Sunstar turned inside out.
- **#17** Kagome Hulls: in the Stella–Jewel and Sunstar lattices, hull-shaped clusters share pieces corner to corner as in the Kagome and pyrochlore nets.
- **#18** The DICTO Hexa: eight exact Jewels passing through each other make one solid, the DICTO Hexa.
- **#38a** The DICTO Kepler–Poinsot–UP Cycle: great and small stellated dodecahedra alternate round the icosahedron in an endless cycle, $\times\varphi^3$ in length each turn.
- **#41** The DICTO Jewel: the Jewel proved exactly (volume 3/16), with the Jewel-Tetra, Jewel-Octa and Jewel-Rhombi as single solids.
- **#42** The DICTO Sunstar: Hart's Dogstar proved exactly (volume $\varphi/2$): dodecahedra and Dogstars fill space; the Sunstar clusters named.

- **#59** The Icosaxes Hexa: Bloom, Bud and Seed: the Hexa move with icosahedral symmetry: twenty Icosaxes close into the Icosaxes Bloom, Bud and Seed.
- **#99** Plato's Five in the Octet: all five Platonic solids nest in every cell of the octet, Pacioli's icosahedron in each octahedron; Sheriffs and Lone-stars fill between icosahedron and dodecahedron.
- **#103** The DICTO Yantra: Two Exact Solutions — Overcoming the Seemingly Impossible, Towards Perfection: the Śrī Yantra on the Cleo-Antonius: Euclid's Cleo-Icos above, a side of Antonius below; the stella octangula as the yantra's frame.

  → #8 → #38a: the nesting I(1) ⊂ GSD ⊂ D(φ) of #38a is the EKP cell nesting of #8 (cross-links audit, row 12).

1.2 The golden cycles

Endless nests of icosahedral solids that repeat φ^3 larger each turn, every layer filled by a few exact pieces.

- **#19** AXE, CLEO and FUJI: AXE, CLEO and FUJI close the thirteen-icosahedron cluster into one icosahedron of edge φ^2; FUJI is new.
- **#20** The FUJI Family: FUJI as one member of a family of pyramids on truncated triangles; Cleo-Icos with Antonius, a space-filling pair from the golden rhombohedron.
- **#22** The Chain of Karma: the icosahedron and the dodecahedron filled exactly as shells of each other, alternating without end.
- **#23** Spheres of Dharma: an exact endless nesting of six icosahedral solids, each layer filled by a few repeated pieces.
- **#24** The Bodhi Seed: a rhombic triacontahedron holding twelve smaller ones and one at its centre, with the Lotus seed’s counts and ratios.
- **#27** Capturing the In-betweens: the in-between stages as solids, among them the DICTO Truncated-Icosahedral Hecatodihedron.
- **#28** The DICTO Hecatodihedron: the 102-faced seventh stage of the Spheres of Dharma.
- **#29** Theaetetus: the hollow DICTO-ICOSA star, its loose core held by Cleo-Icos, grown to the Hecatodihedron.
- **#32** Inverted Shells: The Spheres of Dharma Turned Inside Out: polar reciprocation turns the cycle inside out and reverses every nesting; the Ogdoecontahedron appears.
- **#33** Double Vision – Both Cycles in One Space: both cycles in one space: they share solids on two spheres, never both, because the settings differ by 3, not a power of φ.
- **#34** Progenitors: the twins H and O alone never close on a power of φ (8,742 tight cycles); set into the Spheres of Dharma they close a nine-station golden cycle.
- **#35** Vajrayana: the dodecahedron returns as the hull of the Cleo star, and the loose steps of the cycle are filled.
- **#35a** The DICTO Metaxyhexecontahedron: the solid in the last gap of the golden cycle.
- **#35b** The DICTO Palintriacontahedron: the icosidodecahedron's pyramids flipped out at full height, bridging one turn to the next.
- **#38** Universal PiTet: the growth rule T: from the icosidodecahedron a four-step cycle through the rhombic triacontahedron and the DICTO GTI.
- **#54** One Turn Is a Gnomon: what each step of one turn adds (the Julias and Pimentels); no piece shape repeats at another size.
- **#78** The Truncated Icosahedron Star (TI Star): the truncated icosahedron’s pyramids turned outwards start a golden cycle of their own.
- **#91** Four-Dimensional Solids Whose Every Cell Is a Golden Cut: 119 D·I Golden Tight Cycles close at φ, all through golden-cut stations.
- **#103** The DICTO Yantra: Two Exact Solutions — Overcoming the Seemingly Impossible, Towards Perfection: the DICTO Yantra nests in its own bindu like the gnomon of #54 (ratio 0.0550); its slopes multiply to $2/\varphi^3$.

  → #34 → #91 (+ #35, #33): the same tight-cycle criterion; none of #34’s 8,742 cycles closes on a power of φ, while #91’s golden-cut stations give 119 at φ (row 7).

1.3 OG, the Ogdoecontahedron family

The DICTO Ogdoecontahedron (OG), polar of the Hecatodihedron, with its packings, its relatives up and down, and its place in the octet.

- **#30** A Fibonacci Spine: five body-centred packings of OG set in five wedges round a five-fold axis, every copy at a golden height.
- **#31** The DICTO Ogdoecontahedron: OG: the polar of the Hecatodihedron, 20 regular hexagons and 60 kites, named for its 80 faces.
- **#31a** The DICTO OCD Stella: the hull of the golden adapter pieces round one OG, cut along its 8 joints: the OCD Stella.
- **#34a** The DICTO Hecatogdoecontahedron: the Hecatodihedron and OG interlocked as one self-dual compound with 182 faces and 182 corners.
- **#36** Golden Bodhi Pyrochlore: the 22.57% gap of the OG packing: its exact pieces, the limbs, the GBPC and the Bull-horns adaptor.
- **#37** The DICTO Ogdoecontahedron on the Octet Truss: OG's eight hexagons lie on the faces of the octet's octahedra: four exact space-filling kits.
- **#37a** The DICTO Ogdoecontahedron on the Gyrated Octet: OG in the gyrated octet: the layers are twins by the spinel law; the DBT and the DICTO OTT.
- **#39** The DICTO OG Family: four steps up (rule T) and four down (rule P) from OG: KRE, PHE, TPP, DPHP and TPH, TQH, TQP, TQHD.
- **#40** The DICTO OG Family on Every Stacking: one kit builds the OG family into every c/h stacking, the Fibonacci crystal included.
- **#43** DICTO Kyushindo Cores: every icosahedral solid S lies inside $(15 - 6\sqrt{5})\cdot P(S)$; DICTO's OG turned out is nested at every beat.
- **#80** The Gains of DICTO's Growth Rule: OG's chain under T passes the KRE, PHE, TPP and DPHP, then Hamlet and Ophelia, and does not close.
- **#91** Four-Dimensional Solids Whose Every Cell Is a Golden Cut: OG = $H(\mu TI) \cap \varphi^2 D^*$, and the Lesser and Greater Golden OG at the golden levels.
- **#95** Pentagons for the Kites: DICTO's pentagons for the kites: the cut OG (rectified truncated icosahedron, corners at the C60 bond centres), exact nonzero Dehn invariant, kits for the octet and every stacking (seven shapes), its Bull-horns adaptor, the pyrochlore network.

  → → (4 Nov) OG family fingerprint: icosahedral; W6 sign splits corner-type (TQHD, TQP, OG, KRE) from face-type (TQH, TPH, PHE, TPP, DPHP) in pairs; the RT is nearly silent in degree 6.

1.4 Space-fillers and lattice moves

Single solids and mixtures that fill space, made exactly by moving a lattice, by thirteen-cell blocks or by a kit of connectors.

- **#1** DICTO skewed rhombic dodecahedron: finding: a space-filler of 60° and 72° rhombi, and its two elongated forms.
- **#5** Regular-hexagon elongated dodecahedron: finding: the regular-hexagon elongated dodecahedron, found independently (known: Grünbaum, 2010).
- **#11** The stretched dodecahedron: finding: the dodecahedron stretched along a cube-face axis.
- **#25** DICTO Spheroid-13 RD: a rhombic dodecahedron with its twelve face neighbours fills space by translation alone; DICTO's rounded block.
- **#26** DICTO ATOM-13×13: thirteen polyhedral thirteens side by side, six golden and seven periodic.
- **#45** The Reuleaux Platonic Solids: balls on the corners of all five Platonic solids, as one family: the width theorem.
- **#47** Every Mix of the Platonic Solids Fills Space: every mix of the Platonic solids fills space, with one kit of seven connectors.
- **#49** The Polyhedral Stretching Theorem: a dodecahedron slid one edge-length sweeps an equal-edged solid: the DICTO stretched solids.

- **#50** One Move Each: classical cut-and-glue space-fillers built exactly by one setting of Kaleidohedra's three lattice moves.
- **#93** One Double Pyramid, Two Splits: every caesium chloride pair and every Delian pair except the octet fills space in one way only.
- **#96** Reading the Voids: reading the voids of a packing with spherical waves, harmonics and golden bricks; exact Dehn obstructions for the OG and its adaptors; the DICTO Void Keeper and the DICTO Golden Skin.
- **#99** Plato's Five in the Octet: the nest of all five solids repeats for ever by thirds inside the octet's cells.
- **#100** The Dehn Test Picks the Lone Tilers: a tetrahedron with cut corners fills space alone only when the cuts add up to exactly one edge (the Dehn test).
- **#101** Held by Shape Alone: in an octet layer every octahedron is locked by its neighbours while every tetrahedron lifts straight out; in the Admiral layer only four surface pyramids are free.

1.5 Bonds

Two kinds of point and one sliding weight: space shared out as pairs of convex pieces, and the web of the faces they share.

- **#48** The Golden Split: icosahedra, Unborns and AXE become two convex space-fillers, Icosaxes and Unborn Axes, at the golden point.
- **#51** The Delian Bond: the octet cut by one sliding plane: from rhombic dodecahedra to cubes, with Samson and Delilah at equal volume.
- **#52** The Web of Bonds: each loop is one fixed crowd of points and one weight; the web of bonds is the weighted Delaunay web.
- **#53** The Global Web of Bonds: the weight walked through ten crystal networks; named tilings at the stops.
- **#55** David and Goliath: the A15 stop at 3/4: Goliath is the hull of Jessen’s icosahedron, David a small tetrahedron.
- **#56** Sonsam and Lahdeli: the Delian bond on the twinned octet (Sonsam and Lahdeli); keys change the pieces, never the network.
- **#57** The Rule of Halves: on every stacking each hole's piece is borrowed, whole or in halves, from the two pure stackings.
- **#58** From the Octet to Every Stacking: the Web of Bonds: the series told from the octet to every stacking.
- **#93** One Double Pyramid, Two Splits: the bonds' pairs are rigid: each tiles only as its own crystal (Theorems A, B).

  → #51, #53 → #86: #86’s member A cubes ($\lambda = 3/8$) of edge $1/\sqrt{2}$ are #51’s cube end; member B ($\lambda = -1/8$) is the octet switch (row 9).
  → → (4 Nov) Foam test on the CsCl pairs: wall area largest exactly at Kelvin ($\mu = 3/2$, the swap's fixed point), smallest at CO + O; Delian walk cubic-silent at $t = 1/9$, no icosahedral station.

1.6 Stackings and the Hexagonality Law

Close-packed layers stacked cubic (c) or hexagonal (h) in any order, and the exact laws every stacking obeys, tested on real crystals.

- **#40** The DICTO OG Family on Every Stacking: the tetrahedral–octahedral honeycomb stacks in c/h words, as close-packed crystals do.
- **#57** The Rule of Halves: the Rule of Halves: tetrahedral pieces whole from the nearest layer, octahedral pieces as two halves.
- **#58** From the Octet to Every Stacking: the Web of Bonds: series paper ending in the Rule of Halves.
- **#60** DICTO Janus c/h Pyrochlore: the same rule on the pyrochlore network of diamond-like crystals.
- **#61** The Swap Rule: squashed layers make every piece a Janus piece (Ganesha); the two mixed pieces add up to the two pure ones.
- **#64** The Swap Rule in Crystals: the Swap Rule holds in all 299 testable cases of four crystal families.

- **#66** The Focus: the Hexagonality Law: each kind of piece totals a straight line in the share of hexagonal layers.
- **#67** Random Stackings and Stacking Faults: the law holds for any stacking, faulted or random.
- **#68** Real Polytypes and the Hexagonality Law: in the ANNNI model a property is a line in h exactly when J2 = J3 = 0; silicon carbide data.
- **#69** The Hexagonality Law and the Fibonacci Stacking: the Fibonacci stacking: hexagonal share exactly $1/\varphi^2$, so φ enters the volumes of a crystal.
- **#70** The Hexagonality Law in Real Materials II: entropy and enthalpy under pressure.
- **#71** The Hexagonality Law in Real Materials III: hydrogen sites in titanium and zirconium.
- **#72** The Hexagonality Law in Real Materials IV: Pauling's third rule and Goldschmidt's tolerance factor measured by h.
- **#73** The Hexagonality Law in Real Materials V: ice, diamond and III–V semiconductor nanowires.
- **#74** Spin Ice and Kagome Magnets on Every Stacking: spin ice and kagome magnets on every stacking.
- **#75** HexOct: Two Tiles for Every Stacking: HexOct: only two layer tiles, c and h, build every stacking.
- **#83** HexOct: Knitting Crystals: one plane slices every stacking into a knitting chart.
- **#84** The Window Law: the Window Law: the Hexagonality Law beyond one layer; the Swap Rule is the condition to stay on the line.
- **#86** The HexOct Family: one tower move joins the c-tile; five single-solid tile pairs, A to E.
- **#92** Aristotle's Pyramid in Layers: the face-to-face tilings by the octet's pyramids are exactly the stackings of octet layers.
- **#97** Narcissus Keyed: Narcissus stacks freely; seventeen keyed copies carrying Kari's aperiodic tiles force a sequence of joins that never repeats.
- **#98** Narcissus and Covenant: Narcissus, its own mirror image, fills space alone; two joined on their hexagons give the twisted cube (straight) or Covenant (turned 60°).
- **#102** Every Site Hears the Same: the Loop Equality Is a Theorem: the Loop Equality is a theorem: every site of every close-packed stacking has diamond's closed non-backtracking loops, at every length.
  → #69 → #91 (+ #30): #91's golden-tilt Fibonacci word is #69's c → ch, h → c (share $1/\varphi^2$) (row 10).
  → #67 → #73: #73's ice I_sd value uses #67 Theorem 1 on random stackings (row 11).
  → → (4 Nov) Every stacking is tiled by RD (c) and TRD (h) cells; the harmonic fingerprint reads the c/h word ball by ball; the Fibonacci word is pure point with exact amplitudes while its A/B/C crystal is not (fading peaks at 1/2 and cos 72°); DICTO's turned node = the twisted ball (J27 ties).

1.7 Pyramids, Stars and Vacuus

Cut a solid from its centre into one pyramid per face; then fill space with the pyramids, turn them outwards into stars, or keep only the walls between them.

- **#44** Aristotle's Pyramid: of the five Platonic face pyramids only the cube's alone, or the tetrahedron's with the octahedron's, fill space.
- **#65** The Red Queen 120-cell: all five face pyramids are the cores of the regular four-dimensional solids seen in perspective.
- **#77** Face to Face: A New Law of Aristotle's Pyramids: in face-to-face tilings the pyramids round each apex make up their whole solid.
- **#78** The Truncated Icosahedron Star (TI Star): pyramids turned outwards: the TI Star, its Star Halo, and an endless cycle.
- **#79** Star Cores of DICTO's Solids: the star treatment for DICTO's solids; the pole rule always works; DICTO's valley rule.
- **#80** The Gains of DICTO's Growth Rule: the gains of DICTO's growth rule T found exactly: T is polar reciprocation of the inverted corners.
- **#81** Core Cycles: when the cores of DICTO's star chains alternate: the core turn alternates for every solid whose faces lie in two families of planes.
- **#85** Eccentric Pieces: an edge of dihedral angle δ keeps its valley open through step k only if $(k + 1)\delta \leq 360°$.

- **#87** DICTO Vacuus Solids: keep only the walls between the pyramids: the Vacuus, built as SHEETS and STRUTS.
- **#92** Aristotle's Pyramid in Layers: the octet's pyramid tilings layer by layer, settling the question #77 left open.
- **#92** Aristotle's Pyramid in Layers: version 2: the cuboctahedron's pyramids (regular tetrahedra and half-octahedra) tile as the stackings with a free cut in every octahedron, rebuilding the cuboctahedron on straight layers and the anticuboctahedron on twisted ones.
- **#93** One Double Pyramid, Two Splits: the cube's and the rhombic dodecahedron's centre pyramids: one double pyramid, split across its square into cube pyramids that build only cubes, or along a free diagonal on every square into rhombic dodecahedron pyramids that never close into a cube; #77's law fails for the rhombic dodecahedron.

  → #77 → #92 (+ #40, #57; #51/#86): #77's main theorem is #92's reduction; #92 answers what #77 left open (row 1).
  → #43 → #80, #78, #79, #81 (+ #90, #91): #80's gains $(R/r)^2 = 9, 3, 15 - 6\sqrt{5}$ are #43's constants; #43 Lemma 3 is #80's "T is polar reciprocation" (row 2).
  → #85 → #87: #85's angles $\delta = \alpha + \alpha'$ are the angles of #87's Vacuus fins (row 13).
  → → (4 Nov) Reversed neighbour impossible for T and C6 (search proof): T pyramids form whole tetrahedra, C6 whole cubes, in every tiling; R12 free + slides; the CO pyramids build CO on straight and J27 on twisted layers, and J27 + octahedra tile hcp (twin of the rectified cubic).
  → → (4 Nov) Admiral Cap network (#47) is frustrated (8 of 12 shared edges at most); its ground states are layer words — the same layer freedom as #92.

1.8 The four-dimensional queens

The regular four-dimensional solids, above all the 600-cell (the White Queen) and the 120-cell (the Red Queen): their slices, shadows and the path between them.

- **#62** Shadows of the Fourth Dimension: shadows and slices of the four triangle-built four-dimensional solids: deltahedra of two new kinds.
- **#63** The White Queen 600-cell: a complete exact atlas of the 600-cell's slices in its four main directions; Humpty Dumpty.
- **#65** The Red Queen 120-cell: the 120-cell built exactly from the White Queen; 124 slice patterns.
- **#76** Between the Queens: the five uniform stages between the queens, by cutting the White Queen's corners deeper.
- **#89** All Five Golden Pairs Are Slices of Four-Dimensional Solids: the golden pairs as slices of the tesseract, the 120-cell, the 24-cell and the rectified tesseract.
- **#91** Four-Dimensional Solids Whose Every Cell Is a Golden Cut: DICTO All-Golden Polytopes in the tesseract family and in the 600-cell × 120-cell family.

  → #76 → #91: the three uniform μ of #91's 600 × 120 sweep are #76's edge-equalising cuts (row 3).
  → → (4 Nov) Both Queens are spherical 11-designs; on the path between them the truncated 600-cell at $\mu \approx 0.92417$ is a 19-design (the classical W(H4) orbit, Goethals–Seidel); the 4D sweep's tone zeros: 24-cell, $\sqrt{2}$, $1 + \sqrt{3}$.
  → → (4 Nov) #91's 4D sweep = SO(9) weight shapes; the All-Golden Polytopes are limits of Fibonacci multiplets, as Icosaxes is in 3D (SU(4)).

1.9 The golden cuts

Cut every corner of a solid by the same amount so that every face keeps the golden ratio: where such cuts exist, and where they live.

- **#20** The FUJI Family: FUJI and its family stand on truncated triangles.
- **#48** The Golden Split: the Icosaxes and the Unborn Axes appear, at the golden point of the AXE.
- **#88** Every Platonic Solid Has Exactly Two Golden Cuts: exactly two golden cuts for each Platonic solid; the octahedron's pair is the Icosaxes and the Unborn Axes, which fill space.
- **#89** All Five Golden Pairs Are Slices of Four-Dimensional Solids: all five pairs are slices of four-dimensional solids, the cubes only in the rectified tesseract; beyond Plato golden cuts are rare.
- **#90** Golden on the Octahedron, Silver on the Cube: Six Dimensions and the Golden Cut Cubes: the field obstruction: the Golden Cut Cubes need $\sqrt{2}$ with $\sqrt{5}$; they are Galois twins.

- **#91** Four-Dimensional Solids Whose Every Cell Is a Golden Cut: four-dimensional solids whose every cell is a golden cut.
- **#94** The Silent Golden Station: golden and bronze ratios as the zeros of the cubic harmonics on the sweep.

  → → (4 Nov) Golden cuts under Double Vision: only the cube's pair are Galois twins; the dodecahedron's greater cut (1/3) is its own twin, its lesser cut's twin the φ^3-cut dodecahedron.

1.10 The cube–octahedron sweep

One slider from the octahedron through the cuboctahedron to the cube, met as a weight walk, a polar pair, a six-dimensional section and a four-dimensional path.

- **#48** The Golden Split: the golden network: Icosaxes and Unborn Axes as the two pieces of one sliding pair.
- **#52** The Web of Bonds: the golden loop is the power diagram of the body-centred cubic sites with one sliding weight.
- **#53** The Global Web of Bonds: the caesium chloride (CsCl) and rock-salt lanes of the weight walk.
- **#54** One Turn Is a Gnomon: Theorem B: the polar of the Icosaxes is a cube with six pyramids of height $1/\varphi$.
- **#81** Core Cycles: when the cores of DICTO's star chains alternate: the core turn on $S(\mu) = P \cap \mu P^*$ is an involution.
- **#89** All Five Golden Pairs Are Slices of Four-Dimensional Solids: DICTO's filling by cut cubes and octahedra, a section of the rectified tesseractic honeycomb.
- **#90** Golden on the Octahedron, Silver on the Cube: Six Dimensions and the Golden Cut Cubes: the sections of cube × octahedron sweep from the octahedron through both golden pairs to the cube; cut = shadow picks $\mu = \varphi$.
- **#91** Four-Dimensional Solids Whose Every Cell Is a Golden Cut: $S(\mu) = P \cap \mu P^*$; all ten golden cuts lie on the three sweeps D·I, C·O and T·T.
- **#94** The Silent Golden Station: the harmonic fingerprint of the sweep: the cubic tone Q4 is silent only at Icosaxes ($\mu = \varphi$), whose corners are two icosahedra; Q8 also at the bronze station $(\sqrt{13} - 1)/2$.
- **#96** Reading the Voids: from the harmonic fingerprint (#94) to waves through a whole packing: the resonance line that led to Reading the Voids.

  → #53 → #91 (+ #52): the 24-cell honeycomb sliced vertex-first is #53's rock-salt lane, cell-first the CsCl lane (row 4).

  → #81 → #91, #90: #81's core turn on $S(\mu)$ is $\nu \mapsto (R/r)^2/\nu$; on C·O $\mu \mapsto 3/\mu$, beside the CsCl partner $\mu \mapsto 3 - \mu$ (row 5).

  → #54 → #90, #91: #54 Theorem B is the polar side of the C·O sweep; the rhombic dodecahedron is H(2) (row 6).

  → #53 → #89: DICTO's cut-cube and octahedra filling (#89 §7) is #53's CsCl lane (row 8).

  → → (4 Nov) The sweep is the family of weight shapes of SU(4) ≅ SO(6) then SO(7): octahedron 6 (unison), Kelvin 64 (fifth), cuboctahedron 15/21 (octave), cube 8 (twelfth); golden ladder of Fibonacci multiplets → Icosaxes and Unborn Axes, silver ladder of Pell multiplets → truncated cube (#90's title in particle terms).

  → → (4 Nov) Special points of the sweep by unrelated tests: harmonic Q4 = 0 at φ (Icosaxes), isoperimetric maximum at $\sqrt{3}$ (core-turn fixed point, #81), Mahler maximum near φ (cubic), diamond growth $\alpha = 3/\mu$ (the core turn).

1.11 Mandalas and yantras

Sacred diagrams made exact: the Śrī Yantra's nine triangles meeting three at a time, their solids, their spheres and their exact forms.

- **#103** The DICTO Yantra: Two Exact Solutions — Overcoming the Seemingly Impossible, Towards Perfection: two exact solutions: Kulaichev's classical yantra (degree 348) and the DICTO Yantra (degree 160, the traditional 43); 3D cones, the Cleo sphere of twenty yantras and the 81 sphere (Mandalaspheres in Kaleidohedra).

2. Names

237 entries, alphabetical (a leading “DICTO” is passed over in the order). “Named by” is taken from the paper's credit line or text; “see #N” means the text read for this draft does not say who chose the name; First: the paper or finding (F) where the name first appears; D: the number in the DICTOnary.

Name	Named by	First	D	Also called · note
Admiral Cap	DICTO	F21	D16	icosahedron edge roof
Aetos	DICTO	**#29**	D25	
Alice	DICTO	**#63**	D94	
Alice series slices: Alicia, Unicorn, Bandersnatch	DICTO	**#63**		shapes first found by Alicia Boole Stott (1900)
All-Golden Polytopes, DICTO	DICTO	**#91**		
Antonius	DICTO	**#44**	D65	
ATOM-13×13, DICTO	DICTO	**#26**		
Atraktos	DICTO	**#29**	D24	
AXE	DICTO	**#19**		Mosseri–Sadoc tile t6 (scaled by 1/φ)
B · HexOct −1/8 c-tile	DICTO (#86)	**#86**	D113	whole octahedra
B · HexOct −1/8 h-tile	DICTO (#86)	**#86**	D114	whole octahedra
DICTO Belted Triacontahedron	DICTO	**#37a**	D37	DBT
Blockhead	DICTO	**#56**	D79	
Bodhi seed	DICTO	**#24**		
Bull-horns OG adaptor	DICTO	**#36**	D33	
C · HexOct −1/8 c-tile	DICTO (#86)	**#86**	D115	
C · HexOct −1/8 h-tile	DICTO (#86)	**#86**	D116	
Chain of Karma	DICTO	**#22**		
Chucky	DICTO	**#57**	D80	
CLEO	DICTO	**#19**		Mosseri–Sadoc tile t4 (scaled by 1/φ)
Cleo star	DICTO	**#35**		
Cleo Towers	DICTO	**#43**	D74	8 tower pieces
Cleo-Icos	DICTO	**#29**		Oktaktis (as the OTT’s octahedral ray, #43); the icosahedron’s face pyramid (classical shape)
Crown Jewel	DICTO	**#24**	D17	
Cube Pearl	DICTO	**#80**	D106	
D · HexOct −11/24 c-tile	DICTO (#86)	**#86**	D117	
David	DICTO	**#55**		shape: a tetragonal disphenoid
Delian bond	DICTO	**#51**		
Delilah	DICTO	**#51**		shape: the regular octahedron, at the Delian bond
DICTO Deltoid Aktis	DICTO	**#43**	D55	
Dented tetrahedron L/R	DICTO	**#37**	D72	golden-key octet
DESHI	DICTO	**#19**		
Diadem	DICTO	**#24**		
DICTONIC BONDS	DICTO	**#51**		the three bonds holding the DICTONIC SOLIDS
DICTONIC SOLIDS	DICTO	**#51**		the six convex and keyed solids of #48 and #51
DICTO Dodeca-13	DICTO	**#13**		
DICTO Dodeca-13 Needle	DICTO	**#13**	D7	
DICTO Dodeca-13 Star	DICTO	**#13**	D9	
DICTO Dodeca-13 Unit	DICTO	**#13**	D8	finned unit
DICTO Dodeca-13 Wedge	DICTO	**#13**	D6	
Dodecaxes	DICTO	**#48**	D70	
Dogstar	DICTO	F12		the solid is Hart's stellation 8 of the dodecahedron (1996)
Double Vision	DICTO	**#33**		both cycles in one space
DICTO DPHP	DICTO	**#39**	D48	
E · HexOct 1/10 c-tile	DICTO (#86)	**#86**	D118	
Euclid–Kepler–Pacioli cell and network	DICTO	**#8**		EKP
Fallen Humpty	DICTO (#62)	**#62**	D102	

Name	Named by	First	D	Also called · note
Fang	DICTO	**#57**	D81	
Fibonacci stacking	joint (#40)	**#40**		
DICTO Fill set 1 / 2 pieces	DICTO	**#37a**	D73	grown bipyramid, grown tetrahedra L/R
Five-sided Vajra	DICTO	**#38**	D42	
DICTO FPD	DICTO	**#38**	D41	flat pentakis dodecahedron
frog chorus (five voices)	DICTO	**#30**		A Fibonacci Spine
FUJI	DICTO	**#19**	D12	
Galois twins	DICTO	**#90**		the Greater and Lesser Golden Cut Cubes (D123, D124)
Ganesha	DICTO	**#61**	D84	
GBPC	DICTO	**#36**	D32	Golden Bodhi Pyrochlore Corner
Golden Bodhi Pyrochlore	DICTO	**#36**		
Golden Clasp	DICTO	Star Cores study (2026-10-31)	D107	
Golden OGs: Lesser Golden OG, Greater Golden OG	DICTO	**#91**		
golden tetrahedra (Danzer, Mosseri–Sadoc tiles)	classical	**#38a**		not the Golden Cut Tetrahedra (GCT) of #88/#89: name clash noted in the cross-links audit
Golden TI Star	DICTO	**#78**	D103	
DICTO Golden Tight Cycles	DICTO	**#91**		
DICTO Golden-Regular Cuts	DICTO	**#89**		Greater and Lesser Golden Cut Icosidodecahedron and Snub Dodecahedron
Goliath	DICTO	**#55**		shape: the hull of Jessen's orthogonal icosahedron, the classical Z12 icosahedron of A15
Greater and Lesser Golden Cut Great Dodecahedron, Great Icosahedron, Small and Great Stellated Dodecahedron	DICTO	**#89**		the eight Kepler–Poinsot golden cuts
Greater and Lesser Golden Cut Icosidodecahedron	DICTO	**#89**		found independently; Zefiro 2011's regular-hexagon and regular-decagon stages
Greater Golden Cut Cube	DICTO	**#88**	D123	Galois twin (#90)
Greater Golden Cut Dodecahedron	DICTO	**#88**	D119	
Greater Golden Cut Icosahedron	DICTO	**#88**	D121	
Greater Golden Cut Rhombic Dodecahedron	DICTO	**#89**	D127	
Greater Golden Cut Rhombic Triacontahedron	DICTO	**#89**	D129	
Greater Golden Cut Snub Dodecahedron L/R	DICTO	**#89**	D131	
Greater Golden Cut Tetrahedron	DICTO	**#88**	D125	
DICTO GTI	DICTO	**#38**	D40	Golden Truncated Icosahedron
DICTO-GTT-Solids	DICTO	**#88**		Icosaxes, Unborn Axes, White Knight; two members added in #89
Haigha	DICTO	**#65**	D96	
Hamlet	DICTO	**#80**	D108	
Hatta	DICTO	**#65**	D97	
DICTO Hecatogdoecontahedron	DICTO	**#34a**		
DICTO Hex Aktis	DICTO	**#43**	D54	
DICTO Hexa	DICTO	**#18**		
DICTO Hexa Jewel	DICTO	**#18**	D11	
DICTO Hexa-Key	DICTO	**#18**	D10	
Hexagonal Cleo	DICTO	**#38**	D43	
Hexagonality Law	DICTO	**#66**		
HexOct	DICTO	**#75**		Endless Stack Cell
HexOct c-tile	DICTO	**#75**	D110	λ = 1/10
HexOct family	DICTO (#86)	**#86**		members A–E by weight
HexOct h-tile	DICTO	**#75**	D111	λ = 1/10
HexOct-K	DICTO	**#75**		keyed HexOct tiles
Humpty Dumpty	DICTO	**#63**	D85	
Ian Dury	DICTO	**#56**	D78	
Ian Dury-K, Rhombo-K, Sonsam-K, Lahdeli-K	DICTO	**#56**		the -K (keyed) forms

Name	Named by	First	D	Also called · note
DICTO-ICOSA star	DICTO	**#29**		the DICTO-Star with a pyramid on each of its 152 free faces
Icosaxes	DICTO	**#48**		Greater Golden Cut Octahedron (#88); shape known: Koca et al. 2015, truncated octahedron x = τ
Icosaxes Bloom, Bud and Seed	DICTO (working names)	**#59**		also Lotus Jewel Bloom, Bud, Seed
Icosaxes Hexa	DICTO	**#59**		
DICTO Interlock sets 1 and 2	DICTO	**#33**		Dharma-Ends, Dharma-Middle
Jabberwock	DICTO	**#63**	D88	
Janus c/h pieces	DICTO	**#57**		
DICTO Janus c/h Pyrochlore	DICTO	**#60**		
DICTO Jewel	DICTO	**#41**	D4	Dragon Jewel (first name)
DICTO Jewel-Tetra, Jewel-Octa, Jewel-Rhombi	DICTO	**#41**		first called the DICTO Jewel tetrahedral / octahedral cluster (F15)
Jubjub	DICTO	**#63**	D93	
Julias (G1–G10)	DICTO	**#54**		in tribute to Júlia Pimentel
Kagemusha	DICTO	**#62**	D86	Shadow-17
Kagome Hulls	DICTO	**#17**		
DICTO-Kepler golden truncations	DICTO	**#88**		the ten Greater and Lesser Golden Cut Platonic solids (D119–D126 with Icosaxes and Unborn Axes)
DICTO Kepler–Poinsot–UP Cycle	DICTO	**#38a**		
Keyaxes	DICTO	**#59**	D83	
Keyed octahedron	DICTO	**#37**	D34	golden-key octet
King of Pentacles	DICTO	**#19**	D14	
Kion	DICTO	**#29**	D23	
DICTO KRE	DICTO	**#39**	D45	
Kyushindo series	DICTO	**#43**		
Lahdeli	DICTO	**#56**	D77	
Lesser Golden Cut Cube	DICTO	**#88**	D124	Galois twin (#90)
Lesser Golden Cut Dodecahedron	DICTO	**#88**	D120	
Lesser Golden Cut Icosahedron	DICTO	**#88**	D122	
Lesser Golden Cut Rhombic Dodecahedron	DICTO	**#89**	D128	
Lesser Golden Cut Rhombic Triacontahedron	DICTO	**#89**	D130	
Lesser Golden Cut Snub Dodecahedron L/R	DICTO	**#89**	D132	
Lesser Golden Cut Tetrahedron	DICTO	**#88**	D126	
Lion	DICTO	**#63**	D87	
Lotus Hexa	DICTO	**#59**		
Lotus Jewel	DICTO	F21	D15	clawed icosahedron
Lotus Jewel Axes	DICTO	**#48**	D69	
Lotus seed	DICTO	**#22**		Icosa-13
Mad Hatter	DICTO	**#63**	D91	
Maedate, Kuwagata (notch pieces)	DICTO	**#85**		
March Hare	DICTO	**#63**	D92	
DICTO Metaxyhexecontahedron	DICTO	**#35a**	D30	M
Mirror middle piece Mg	DICTO	**#37a**	D39	
Monkey Staff	DICTO	**#27**	D20	
Monkey Staff B	DICTO	**#38**	D44	
Mono-Strut, Mono-Length, Disc and Isofin Vacuus	DICTO	**#87**		
DICTO OCD Stella	DICTO	**#31a**	D29	
OG turned out	DICTO	**#43**		
DICTO Ogdoecontahedron	DICTO	**#31**	D28	OG; OCD; O
Oktaktis	DICTO	**#43**		the Cleo-Icos as the OTT's octahedral ray
Ophelia	DICTO	**#80**	D109	
DICTO OTT	DICTO	**#37a**	D38	Octahedrally Truncated Triacontahedron
DICTO Palintriacontahedron	DICTO	**#35b**	D31	

Name	Named by	First	D	Also called · note
Penta-cleo	DICTO	**#20**	D66	
DICTO PHE	DICTO	**#39**	D46	
Pimentels (G11)	DICTO	**#54**		in tribute to Júlia Pimentel
PiTet (Universal PiTet)	DICTO	**#38**		Pi for the spherical progression, Tet for the triangles
Polar Cleo Towers	DICTO	**#43**	D75	7 tower pieces + core ray
Red King	DICTO	**#65**	D98	
Red Knight	DICTO	**#65**	D100	
Red Queen	DICTO	**#65**		the 120-cell
Regular-hexagon elongated dodecahedron	known (Grünbaum 2010); found independently by DICTO	F5		
Reuleaux dodecahedron, icosahedron, cube and octahedron	DICTO (names by look)	**#45**		not Reuleaux polyhedra in the strict sense (#45)
Ritsudō cycle	DICTO	**#23**		the Spheres of Dharma nesting
Rule of Halves	DICTO	**#57**		
Samson	DICTO	**#51**	D71	
Shield	DICTO	**#28**	D21	
DICTO skewed elongated dodecahedron ED F16	DICTO	F7	D2	
DICTO skewed elongated dodecahedron ED F18	DICTO	F7	D3	
DICTO skewed rhombic dodecahedron	DICTO	F1	D1	
Slice-48, 12-dented icosahedron, augmented dodecahedron	DICTO (#62)	**#62**		
Sonsam	DICTO	**#56**	D76	
Sphen	DICTO	**#29**	D26	
Sphenion	DICTO	**#29**	D27	
Spheres of Dharma	DICTO	**#23**		
DICTO Spheroid-13 RD	DICTO	**#25**		
Splash Rules	DICTO	**#61**		rule finder, 27 October 2026 (#84)
DICTO-Star	DICTO	**#14**		
Star Chain Reaction	DICTO	F12		told in #42
Star Halo	DICTO	**#78**	D104	
Stella–Jewel Lattice	DICTO	**#17**		
Stretched dodecahedron	DICTO	F11	D5	
DICTO stretched solids (equal-edged, telescoped)	DICTO	**#49**		
Stupa	DICTO	**#27**	D19	
Sunstar	DICTO	F12		a dodecahedron with the Dogstars of its six face-neighbour cells (#42)
Sunstar Lattice, Sunstar-Tetra, Sunstar-Octa, Sunstar-Rhombi, Sunstar-Hexa	DICTO	**#42**		Sunstar clusters (F16)
DICTO Sunstar-Hexa-Key	DICTO	**#42**	D53	
Swap Rule	DICTO	**#61**		
Tetartoid core	DICTO	**#37**	D36	golden envelope network
DICTO Tetraktis	DICTO	**#43**	D57	OTT rhombic ray
Theaetetus (world)	DICTO	**#29**		after Theaetetus of Athens
TI Star	DICTO	**#78**		Truncated Icosahedron Star; Golden TI Star (D103) at s = φ
TI Star core	DICTO	**#78**	D105	
DICTO TPH	DICTO	**#39**	D49	Triangle-Pentagon Hecatontatriacontadihedron
DICTO TPP	DICTO	**#39**	D47	
DICTO TQ-320	DICTO	**#43**	D59	
DICTO TQ-500	DICTO	**#43**	D61	
DICTO TQD-732	DICTO	**#43**	D62	
DICTO TQH	DICTO	**#39**	D50	Triangle-Quadrilateral-Hexagon Polyhedron
DICTO TQHD	DICTO	**#39**	D52	Triangle-Quadrilateral-Hexagon Diacosihedron

Name	Named by	First	D	Also called · note
DICTO TQHD-662	DICTO	**#43**	D63	
DICTO TQP	DICTO	**#39**	D51	Triangle-Quadrilateral-Pentagon Polyhedron
DICTO TQP-252	DICTO	**#43**	D58	
DICTO TQP-402	DICTO	**#43**	D60	
DICTO TQP-672	DICTO	**#43**	D64	
DICTO Triaktis	DICTO	**#43**	D56	OTT triangle ray
Tricap	classical	**#48**		Mosseri–Sadoc tile t3
Triskelion Tetrahedron	DICTO	**#57**	D82	TT
DICTO Truncated-Icosahedral Hecatodihedron	DICTO	**#28**	D22	DICTO Hecatodihedron; H
Tweedledee	DICTO	**#63**	D90	
Tweedledum	DICTO	**#63**	D89	
Twisted cube	DICTO (#86)	**#86**	D112	A · HexOct 3/8 h-tile
Unborn	DICTO	**#48**		the squeezed icosahedron in the cubic grid of icosahedra
Unborn Axes	DICTO	**#48**		Lesser Golden Cut Octahedron (#88); shape known: Koca et al. 2015, truncated octahedron x = 1/τ
UNITY, VENUS	DICTO	**#19**		UNITY: icosidodecahedron + 12 J11; VENUS its closed form (#26)
DICTO Vacuus Solids	DICTO	**#87**		after Leonardo's vacuus bodies in Pacioli's De divina proportione
VAJRA	DICTO	**#19**	D13	
Vajra (fine)	DICTO	**#24**	D18	
W	DICTO	**#37**	D35	one-piece glue
Walrus	DICTO	**#65**	D99	
White Knight	DICTO	**#65**	D101	
White Queen	DICTO	**#63**		the 600-cell
White Rabbit	DICTO	**#63**	D95	
Window Law	DICTO	**#84**		
X	DICTO	**#47**	D67	cube of edge φ minus one Euclid roof (#47 connector)
Y	DICTO	**#47**	D68	cube of edge φ plus one Euclid roof (#47 connector)
Silence Is Golden	DICTO	Golden Ladder ch. 2		chapter title: the golden stations are where the cubic tone falls silent
Golden Silences	DICTO	app (planned)		app world: hear the sweep as a tuning instrument; the cubic tone falls silent at the golden stations (#94, Golden Ladder ch. 2)
Narcissus	DICTO	HexOct: Narcissus and Covenant		
Echo	DICTO	HexOct: Narcissus and Covenant		
Covenant	DICTO	HexOct: Narcissus and Covenant		
DICTO Void Keeper	DICTO	**#96**	D133	
DICTO Golden Skin	DICTO	**#96**	D134	
DICTO Void set 1 (cut OG), DICTO Void set 2 (OG)	DICTO	**#96**		
Sheriff	DICTO	**#99**		pentagonal cap of I − D: a corner of the icosahedron over a face of its dual dodecahedron; slant edges √3/φ × base
Lone-star	DICTO	**#99**		wedge of I − D: an icosahedron edge and the dodecahedron edge beneath it, crossing at right angles; four equal edges
DICTO Yantra	DICTO	**#103**		the Śrī Yantra on the Cleo-Antonius pair that meets Kulaichev's three conditions with the traditional 43; exact minimal polynomial of degree 160 (DICTO, 8 Nov 2026): great pair Cleo-Icos and Antonius side, slope product 2/φ³; both small triangles on the circle, bindu centred, 43 triangles, 18 marmas, 24 sandhis (study GF1). Earlier the name stood for K1, the Cleo-Octangula kite.
Cleo-Octangula	DICTO	**#103**		the near-traditional variant: great pair Cleo-Icos above and stella tetrahedron below (slopes 2/φ², 1/√2); its yantra K1 and, from the perfect viewpoint, V1

Name	Named by	First	D	Also called · note
Cleo-Antonius	DICTO	**#103**		the great pair of the DICTO Yantra: the Cleo-Icos (cone slope $2/\varphi^2$) with a side of Antonius (slope $1/\varphi$); product $2/\varphi^3$
DICTO Hopscotch	DICTO	outline The DICTO Hopscotch (9 Nov 2026)		the unifying framework: polar pairs along a slider, events where one determinant vanishes; DICTO predicted the common denominator (9 Nov 2026)
Antipol	DICTO	outline The DICTO Hopscotch (9 Nov 2026)		a shape together with its polar, facing each other; a square of the Hopscotch
hop (Hopscotch)	DICTO	outline The DICTO Hopscotch (9 Nov 2026)		from a value to the same value: zero to zero along a slider, or a return around a cycle
step (Hopscotch)	DICTO	outline The DICTO Hopscotch (9 Nov 2026)		hand-over with emergence before dissolution (gradual; #53 Proposition D, tau_A + tau_B < 0)
jump (Hopscotch)	DICTO	outline The DICTO Hopscotch (9 Nov 2026)		emergence and dissolution at one value (switch; #53 Theorem B, tau_A + tau_B = 0)
turn (Hopscotch)	DICTO	outline The DICTO Hopscotch (9 Nov 2026)		the end of the line where one turns and comes back: the polarity's fixed point ($\sqrt{3}$ for $\mu \mapsto 3/\mu$)
emergence / dissolution	DICTO	outline The DICTO Hopscotch (9 Nov 2026)		the appearing of new bonds and the disappearing of old ones along a slider

3. Almanac of Concepts

44 concepts: operations, constructions, laws, theorems and methods, as distinct from the named shapes. Each definition is taken from the wording of the paper cited; then the first paper and its date, the papers that use it, its threads, and who proposed it.

c/h stacking words. Each layer sits on the one below as in the octet (c) or as its mirror image (h); any word in c and h is a stacking (#40).

First: **#40** The DICTO OG Family on Every Stacking (20 October 2026) · Used in: #57, #58, #60, #61, #64, #66, #67, #74, #83, #84, #86, #92 · Threads: Stackings and the Hexagonality Law; OG, the Ogdoecontahedron family · Proposed: Jagodzinski's notation (classical); taking the OG family to every stacking was proposed by Claude and adopted by DICTO (#40)

core turn (core cycle). The solid common to the hulls of the tips of each family of faces; for the truncated icosahedron it alternates, and two turns give it back (#81).

First: **#78** The Truncated Icosahedron Star (TI Star) (current version 3 November 2026) · Used in: #81, #90, #91 · Threads: Pyramids, Stars and Vacuus; The cube–octahedron sweep · Proposed: DICTO found it for TI (#78) and proposed it for every two-family solid (#81)

cut equals shadow. Asking the cut to equal the shadow of a lattice orbit selects $\mu = \varphi$, where the roots of D6 cast the Icosaxes point for point (#90).

First: **#90** Golden on the Octahedron, Silver on the Cube: Six Dimensions and the Golden Cut Cubes (4 November 2026) · Used in: — · Threads: The cube–octahedron sweep; The golden cuts · Proposed: see #90

Dehn invariant (counting). A polyhedron that tiles space has Dehn invariant zero; #44 makes a Dehn-type count on dihedral angles rigorous with the Splitting Theorem of Conway, Radin and Sadun.

First: **#22** The Chain of Karma (current version 20 October 2026) · Used in: #36, #37, #37a, #38, #38a, #44 · Threads: Pyramids, Stars and Vacuus; The golden cycles · Proposed: classical (Dehn; Debrunner)

Delian bond. The cut of the octet's double cones at which the two solids have the same volume: the tetrahedron's has doubled and the octahedron's halved (#51).

First: **#51** The Delian Bond (24 October 2026) · Used in: #52, #53, #56, #85, #92 · Threads: Bonds · Proposed: DICTO proposed the approach and named Samson and Delilah; doubling is the ancient Delian problem (#51)

DICTONIC SOLIDS and BONDS. The six convex and keyed solids of #48 and #51, held in three bonds (#51).

First: **#51** The Delian Bond (24 October 2026) · Used in: #52, #53 · Threads: Bonds · Proposed: DICTO (names, #51)

EKP construction. Put Euclid's six roofs on a cube and reflect the roof corners back through the cube faces: they land on a regular icosahedron inside (#8).

First: **#8** The Euclid–Kepler–Pacioli Cell (current version 20 October 2026) · Used in: #17, #18, #41, #42, #79 · Threads: Kepler, Euclid and the stars · Proposed: DICTO's construction (#8)

face-to-face reduction. In every face-to-face tiling by face pyramids, the pyramids round each apex make up their whole solid (#77).

First: **#77** Face to Face: A New Law of Aristotle's Pyramids (30 October 2026) · Used in: #92 · Threads: Pyramids, Stars and Vacuus; Stackings and the Hexagonality Law · Proposed: DICTO proposed the face-to-face case (#77)

Fibonacci stacking. Layers c and h in the never-repeating order of c → ch, h → c; the hexagonal share is exactly $1/\varphi^2$ (#69).

First: **#40** The DICTO OG Family on Every Stacking (20 October 2026) · Used in: #69, #83, #84, #91 · Threads: Stackings and the Hexagonality Law; OG, the Ogdoecontahedron family · Proposed: joint work (#40)

field obstruction. The Golden Cut Cubes need $Q(\sqrt{2}, \sqrt{5})$, so no point set with coordinates in $Q(\sqrt{5})$ holds a similar copy of one (#90).

First: **#90** Golden on the Octahedron, Silver on the Cube: Six Dimensions and the Golden Cut Cubes (4 November 2026) · Used in: — · Threads: The golden cuts; The cube–octahedron sweep · Proposed: see #90

gnomon. The piece which, added to a figure, makes a whole of the same shape, only larger (Euclid, Heron) (#54).

First: **#54** One Turn Is a Gnomon (25 October 2026) · Used in: #66, #78 · Threads: The golden cycles · Proposed: classical; DICTO's idea to carry it into his three-dimensional golden cycle (#54)

golden cut. Cutting every corner of a solid equally so that each face becomes a truncated polygon with its sides in the golden ratio φ (#88).

First: **#88** Every Platonic Solid Has Exactly Two Golden Cuts (current version 3 November 2026) · Used in: #89, #90, #91 · Threads: The golden cuts; The cube–octahedron sweep · Proposed: DICTO's observation of the 3-fold hexagon in his solids; the solids and family names are DICTO's (#88)

golden cycle. An exact, endlessly repeating nesting of icosahedral solids, each layer filled by a few repeated pieces, growing by φ^3 each turn (#22, #23).

First: **#22** The Chain of Karma (current version 20 October 2026) · Used in: #23, #24, #32, #33, #34, #35, #54, #78, #91 · Threads: The golden cycles · Proposed: DICTO's (#22, #23)

golden split. Cut each AXE into four little pyramids from one inner point and give each to the piece it touches; at 1/φ along its axis nothing sticks out (#48).

First: **#48** The Golden Split (23 October 2026) · Used in: #51, #52, #59 · Threads: Bonds; The golden cuts · Proposed: DICTO: pieces without odd protrusions, cuts "hemispherical, or anywhere in between" (#48)

growth rule T. Every corner v of a solid gives the face plane v·x = v·v, through v and at right angles to the line from the centre; T(S) is the solid these planes bound (#39).

First: **#38** Universal PiTet (current version 20 October 2026) · Used in: #39, #43, #78, #79, #80, #81, #91 · Threads: The golden cycles; OG, the Ogdoecontahedron family; Pyramids, Stars and Vacuus · Proposed: DICTO's growth rule (#80); T is the 3D form of the classical antipedal construction (#38 v3)

Hexagonality Law. In every stacking each kind of piece totals exactly what its layers hold in their own pure stackings, a straight line in the share of hexagonal layers (#66).

First: **#66** The Focus (current version 28 October 2026) · Used in: #67, #68, #69, #70, #71, #72, #73, #74, #75, #83, #84, #90, #91 · Threads: Stackings and the Hexagonality Law · Proposed: DICTO's law (#84)

HexOct layer tiles. The tile of a layer is its share of space, four pieces; only two tiles, c and h, occur in any stacking (#75).

First: **#75** HexOct: Two Tiles for Every Stacking (current version 29 October 2026) · Used in: #83, #86 · Threads: Stackings and the Hexagonality Law · Proposed: DICTO's idea and name (#75)

Janus c/h pieces. Octahedral pieces with one cubic and one hexagonal half, looking into one stacking above and the other below (#57).

First: **#57** The Rule of Halves (26 October 2026) · Used in: #58, #60, #61, #64, #66, #67, #74, #83 · Threads: Stackings and the Hexagonality Law · Proposed: DICTO (name, #57)

Kagome hull test. Group cells into hull-shaped clusters and ask whether any two share at most one piece and every shared piece lies in exactly two (#17).

First: **#17** Kagome Hulls (current version 20 October 2026) · Used in: #18, #41, #42, #59 · Threads: Kepler, Euclid and the stars · Proposed: DICTO's test (#17)

keys and dents. A key or a dent moves small pieces from one solid to its partner; keys change the pieces but never the network (#54, #56).

First: **#37** The DICTO Ogdoecontahedron on the Octet Truss (current version 20 October 2026) · Used in: #48, #51, #54, #56, #57, #58, #59, #75 · Threads: Bonds; OG, the Ogdoecontahedron family · Proposed: DICTO (keyed glue, #37; keyed versions of the new pairs, #56)

knitting slice. One plane, perpendicular to a row of touching balls, cuts every tetrahedron into an isosceles triangle and every octahedron into a rhombus, a chart to knit (#83).

First: **#83** HexOct: Knitting Crystals (current version 1 November 2026) · Used in: — · Threads: Stackings and the Hexagonality Law · Proposed: DICTO proposed the idea (#83)

lattice moves. A shear of the lattice, the equal-edged cell and a stretch rebuild classical space-fillers in one setting (#50).

First: **#50** One Move Each (23 October 2026) · Used in: — · Threads: Space-fillers and lattice moves · Proposed: Fedorov's moves (1885); DICTO's idea that they rebuild the cut-and-glue fillers (#50)

polar reciprocation. Replace each solid by its polar reciprocal: the nesting of a cycle reverses (#32).

First: **#32** Inverted Shells: The Spheres of Dharma Turned Inside Out (current version 20 October 2026) · Used in: #33, #34, #38, #43, #54, #80, #87, #91 · Threads: The golden cycles; OG, the Ogdoecontahedron family; Pyramids, Stars and Vacuus; The cube–octahedron sweep · Proposed: classical; turning the cycle inside out by it was DICTO's idea (#32)

polyhedral stretching. Slide a solid in a straight line and take the shape it sweeps through (#49).

First: **#49** The Polyhedral Stretching Theorem (22 October 2026) · Used in: — · Threads: Space-fillers and lattice moves · Proposed: DICTO (the stretched dodecahedron, F11, #49)

pyramidise (centre pyramids). Split a solid whose centre lies inside every face plane into its centre pyramids, one cone from the centre over each face (#44, #87).

First: **#44** Aristotle's Pyramid (21 October 2026) · Used in: #65, #77, #78, #79, #81, #85, #87, #92 · Threads: Pyramids, Stars and Vacuus · Proposed: Aristotle's pyramid (classical); DICTO proposed pyramidise as one operation on any such solid (#87)

Reuleaux construction. A ball on every corner of a Platonic solid, radius the solid's diameter; the body is what all the balls share (#45).

First: **#45** The Reuleaux Platonic Solids (21 October 2026) · Used in: #87 · Threads: Space-fillers and lattice moves · Proposed: DICTO's idea (#45)

reverse step P. The step back: take as new corners the foot points of the face planes, the points of those planes nearest the centre (#39).

First: **#39** The DICTO OG Family (20 October 2026) · Used in: #43, #80 · Threads: OG, the Ogdoecontahedron family · Proposed: DICTO proposed running the rule the other way (#39)

rooms and windows. Reading the wave field like the electron density of a crystal: peaks are atoms, saddles bonds and rings (windows), minima cages (rooms); atoms − bonds + rings − cages = 0 (#96).

First: **##96** (date to check) · Used in: #96 · Threads: Space-fillers and lattice moves · Proposed: DICTO's idea (#96)

Rule of Halves. A tetrahedral gap takes the piece of the pure stacking of its nearest layer; an octahedral gap's piece is two halves, from the layers above and below (#57).

First: **#57** The Rule of Halves (26 October 2026) · Used in: #58, #60, #61, #64, #66, #67, #69, #72, #74, #75, #83, #84, #85, #86, #92 · Threads: Stackings and the Hexagonality Law; Bonds · Proposed: DICTO (#84 credit); DICTO's idea to follow the web onto every stacking (#57)

shadow forms. Other ways of breaking up the same space, to find new pieces (#48).

First: **#48** The Golden Split (23 October 2026) · Used in: #51 · Threads: Bonds · Proposed: DICTO's idea (#48)

spherical-wave reading. Every point of solid sends out a wave sin(kr)/(kr); at the packing's own wavelength the waves cancel in the gaps, so the dark places are the voids (#96).

First: **##96** (date to check) · Used in: #96 · Threads: Space-fillers and lattice moves; The cube–octahedron sweep · Proposed: DICTO's idea (#96)

spinel twin law. Twinning on an octahedron-face plane by a half-turn about ⟨111⟩; for OG the twin is OG turned 60° (#37a).

First: **#36** Golden Bodhi Pyrochlore (current version 20 October 2026) · Used in: #37a, #40 · Threads: OG, the Ogdoecontahedron family; Stackings and the Hexagonality Law · Proposed: classical crystallography

Splash Rules. Look for one simple rule behind a whole family of exactly computed values (#84).

First: **#61** The Swap Rule (current version 27 October 2026) · Used in: #84 · Threads: Stackings and the Hexagonality Law · Proposed: DICTO, 27 October 2026 (#84)

Star and Star Halo. Turn a solid’s face pyramids outwards and it becomes a star; the tightest skin round the tips is the star’s halo (#79).

First: **#78** The Truncated Icosahedron Star (TI Star) (current version 3 November 2026) · Used in: #79, #80, #81, #85 · Threads: Pyramids, Stars and Vacuus; The golden cycles · Proposed: DICTO's idea (#78); the name Star Halo is DICTO's

Swap Rule. The two mixed pieces always add up to the two pure ones (#61).

First: **#61** The Swap Rule (current version 27 October 2026) · Used in: #64, #66, #67, #69, #72, #74, #84 · Threads: Stackings and the Hexagonality Law · Proposed: DICTO (name, #61)

the sweep S(μ) = P ∩ μP*. The flats y = x/μ of the product P × P* of a solid and its polar cut out P ∩ μP*; every corner cut of P or P* is such a section (#91).

First: **#90** Golden on the Octahedron, Silver on the Cube: Six Dimensions and the Golden Cut Cubes (4 November 2026) · Used in: #91 · Threads: The cube–octahedron sweep; The golden cuts · Proposed: DICTO proposed cube × octahedron as the six-dimensional home of the pair (#90) and following each solid with its dual through the dimensions (#91)

tight cycles. Nestings in which every step is tight; one that closes at a power of φ is a DICTO Golden Tight Cycle (#34, #91).

First: **#34** Progenitors (current version 20 October 2026) · Used in: #91 · Threads: The golden cycles · Proposed: see #34; the name DICTO Golden Tight Cycles is DICTO’s (#91)

tower move. Moving one tower of the c-tile by one lattice step along its own layer joins the towers along a face (#86).

First: **#86** The HexOct Family (3 November 2026) · Used in: — · Threads: Stackings and the Hexagonality Law · Proposed: DICTO proposed true simple pieces (#86)

truncated triangle. A hexagon with all six angles 120° and sides alternating between two lengths; the regular hexagon is the case of equal sides (#88).

First: **#20** The FUJI Family (date to check) · Used in: #88, #89, #91 · Threads: The golden cuts · Proposed: classical face; DICTO observed that it runs through his solids (#88)

Vacuus: SHEETS and STRUTS. Keep only the walls between the centre pyramids, a fin O–A–B for each edge; built as SHEETS (the fins) and STRUTS (a rod from the centre to every corner, with the edges) (#87).

First: **#87** DICTO Vacuus Solids (3 November 2026) · Used in: — · Threads: Pyramids, Stars and Vacuus · Proposed: DICTO's idea and name (#87)

valley rule. Shorten each spike just enough to clear its neighbours: the ridges keep full spikes and the valleys flatten (#79).

First: **#79** Star Cores of DICTO's Solids (current version 1 November 2026) · Used in: #85 · Threads: Pyramids, Stars and Vacuus · Proposed: DICTO's rule (#79)

web of bonds. Draw one line, a bond, between every two pieces that share a face; the lines make a web, the classical weighted Delaunay web (#52).

First: **#52** The Web of Bonds (25 October 2026) · Used in: #53, #56, #57, #58 · Threads: Bonds · Proposed: DICTO's idea, 25 October 2026 (#52)

weight walk (power cell). Each piece is the region nearer, in a weighted sense, to its own point than to any other; one number, a weight given to one kind of point against the other, slides (#52, #53).

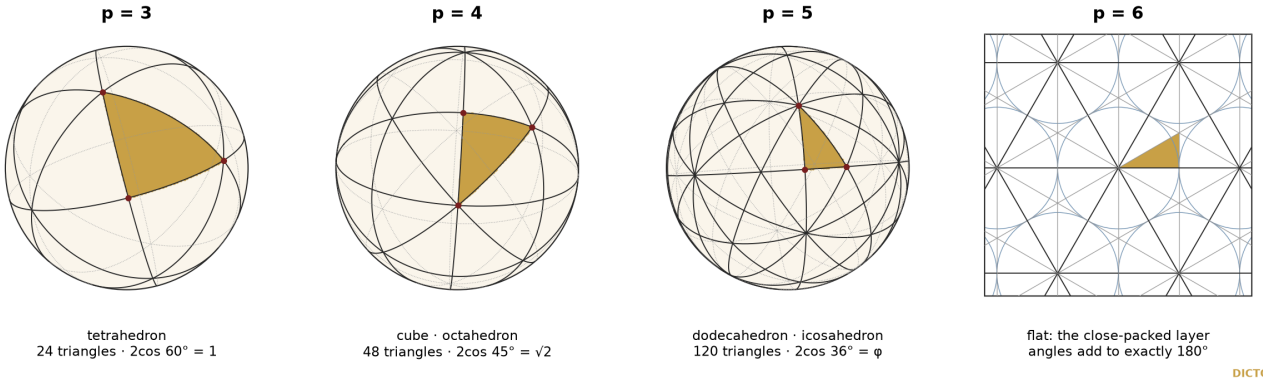
First: **#52** The Web of Bonds (25 October 2026) · Used in: #53, #55, #56, #57, #58, #60, #61, #64, #66, #67, #69, #75, #86, #90, #91 · Threads: Bonds; Stackings and the Hexagonality Law; The cube–octahedron sweep · Proposed: power diagrams are classical (Aurenhammer); DICTO's two loops read this way (#52)

Window Law. Anything fixed by g consecutive layers has a mean that is linear in the shares of the c/h words of length g (#84).

First: **#84** The Window Law (1 November 2026) · Used in: — · Threads: Stackings and the Hexagonality Law · Proposed: DICTO's idea (#84)

4. The Singularity: One Triangle

The Golden Ladder, chapter 1 (One Triangle, Three Numbers; The Golden Ladder, DICTO DOI: 10.5281/zenodo.23966554) shows that the five Platonic solids come from three kaleidoscopes, the spherical triangles with angles 90°, 60° and 180°/p for p = 3, 4, 5, and that each carries one number, 2 cos(180°/p): φ for the dodecahedron and icosahedron (p = 5), √2 for the cube and octahedron (p = 4), 1 for the tetrahedron (p = 3). At p = 6 the triangle is flat and tiles the plane under the close-packed layer of balls, the setting of the stacking papers. The table places each thread by its number world and by the dimensions its papers reach.



The (2, 3, p) triangle for p = 3, 4, 5 on the sphere and p = 6 in the plane, with the number 2 cos(180°/p) of each (figure 1 of The Golden Ladder, chapter 1).

Thread	Number world	2D	3D	4D	6D	8D	From
Kepler, Euclid and the stars	φ (p = 5), on the cube		●				Euclid's cube with roofs; the Dogstar ties the cube to the dodecahedron (ch. 1 §4)
The golden cycles	φ (p = 5)		●	●			icosahedral nests ×φ³; Lotus seed centres = 600-cell vertex and neighbours (#22); #91
OG, the Ogdoecontahedron family	φ (p = 5); flat (p = 6) in #40		●		●		OG in Q(√5); OG as section and hull of the six-dimensional product (#91)
Space-fillers and lattice moves	1, √2 and φ		●				all five solids and their mixes (#47)
Bonds	√2 (p = 4), φ in the golden loop	●	●				octet and cube (#51); icosahedra (#48); three plane networks (#53)
Stackings and the Hexagonality Law	flat (p = 6); φ in the Fibonacci stacking	●	●				close-packed layers; share 1/φ² (#69)
Pyramids, Stars and Vacuus	1, √2 and φ		●	●			T + O and cube pyramids (#44); TI Star (#78); pyramids as cores of 4D solids (#65)
The four-dimensional queens	φ (p = 5)		●	●			600-cell and 120-cell, numbers a + b√5 (#63)
The golden cuts	1, √2 and φ	●	●	●	●	●	c_n = 2cos(180°/n) (#88); 4D (#89); Z6 and E8 → H4 (#90)
The cube–octahedron sweep	√2 (p = 4), golden stations		●	●	●		C·O sweep; rectified tesseract (#89); cube × octahedron (#90)

The golden cuts reach furthest, to the E8 lattice in eight dimensions (#90); OG and the cube–octahedron sweep reach six (#90, #91). The flat triangle (p = 6) carries the stackings, where φ enters through the Fibonacci stacking (#69). The placement is the compiler's reading of the papers named in the last column.

5. What Is Missing

> *Parts 1–4 hold what we know. This part holds what we do not: gaps inside the work, the open points the papers left, and, from other fields, results we did not know of that bear on ours. DICTO's point: the Concordance, the reference library and the Kaleidohedra engine together are the tool. Each entry is to carry three legs, the map (threads and papers it touches), the source (a library copy) and the engine (a view or test in the app); a missing leg is itself a gap (draft 2 carries the library leg for the outside finds; the engine leg follows), and each outside find moves from to read to credited, tested, and finally ours.*

5.1 The grid: solids against operations

Papers where each operation is done for each solid; — marks a gap, ? an unsure entry. PYR centre pyramids and their tilings · STAR Star and Star Halo · VAC Vacuus · GC golden cut · SWP polar sweep P ∩ μP* · NEST nesting and golden cycles · TWIN Galois twin (Double Vision) · CH c/h stackings · DEHN Dehn count · 4D four-dimensional lift or slice.

Solid	PYR	STAR	VAC	GC	SWP	NEST	TWIN	CH	DEHN	4D
Dodecahedron	44, 65	35, 78, 80	87	88, 89	91	22, 23, 33, 34, 91	33, 38a	—	22, 44	65, 89, 91
Icosahedron	44, 65	78, 80	87	88, 89	91	19, 22, 23, 38a, 91	33, 38a	—	22, 44	63, 65, 89, 91
Cube	44, 65, 93	80, 81, 87	87	88, 89, 90	90, 91	8, 91	90	86 (?)	44	65, 89, 90, 91
Tetrahedron	44, 77, 92	80, 81	87	88, 89	91	8, 91	—	40, 57–75, 92	44	65, 89, 91
Octahedron	44, 77, 92	80	87	48, 88, 89	90, 91	8, 91	—	40, 57–75, 92	44	65, 89, 90
Rhombic dodecahedron	93	80, 87	87	89	station only	91	—	—	—	91 (classical)
Cuboctahedron	—	80	87	none exists (89)	station only	?	—	—	—	91 (?)
Rhombic triacontahedron	—	38, 80	87	89	station only	23, 24, 38, 91	33 (?)	—	38 (?)	—
Icosidodecahedron	35b (star)	38, 80	87	golden-regular only (89)	station only	23, 38, 91	33 (?)	—	—	63, 89

5.2 The nearest gaps, ranked, inside and outside

Ranked by how directly each follows from what is already proved (DICTO: most of the calculations are already done; the rest is arrangement and interpretation). DICTO's observation: the gaps inside the work are directly analogous to the gaps outside it, so it is one study. Each gap therefore names the outside work that bears on it (5.5); a gap with none is where to search outside next.

1. **The cuboctahedron's centre pyramids**. Why near: its circumradius equals its edge, so its pyramids are regular tetrahedra and half-octahedra, the pieces #92 classified; #77's reduction applies directly (settled face to face on 4 November 2026: #92 version 2, §5) — published in #92 v2, Theorem 10. Leads to: a third member beside #92 and #93. Outside: *none found yet: a search target.*
2. **The second dual sweeps, cuboctahedron–rhombic dodecahedron and icosidodecahedron–triacontahedron**. Why near: #91 proves P ∩ μP* holds every corner cut, so #89's golden rhombic cuts must sit on these sweeps — BEGUN 4 Nov

2026 (studies/2026-11-04-co-rd-sweep): CO–RD sweep face census; harmonic Q4 zero once at a non-golden point. Leads to: the next rung of the dual ladder; tight cycles through the rhombic solids. Outside: C. Wild et al. … J. Phys.: Condens. Matter 36 (2024) 045303.

3. **The reversed neighbour (#92 open point 1, #93 open point 1)**. Why near: the same local question for the tetrahedron pyramid and the cube pyramid; one argument may close both — SETTLED 4 Nov 2026 (studies/2026-11-04-reversed-neighbour): impossible for T and C6, search proof with overlap margin 0.20; T pyramids form whole tetrahedra, C6 whole cubes, in every tiling; to #92 v3, #93 v2. Leads to: every tiling falls into whole solids, settling #44 for all tilings. Outside: M. Goldberg … Discrete Comput. Geom. 37 (2007).
4. **#91 Lemma 6 exact**. Why near: #76's exact constructions are the stations needed (cross-link). Leads to: #91's 600 × 120 result as a theorem. Outside: T. Denney et al. … Canad. Math. Bull. 28 (1985) 385.
5. **#43 Conjecture 2 (octahedral constant 3)**. Why near: #80's gain 3 and #81's involution μ ↦ 3/μ give the same constant — PROVED 8 Nov 2026 (studies/2026-11-08-octahedral-constant; published in #43 v2, Theorem 4, Zenodo 24052506). Leads to: an octahedral twin of #43 Theorem 2. Outside: *none found yet: a search target*.
6. **The Hexagonality Law for the Laves big atom (#66, #67)**. Why near: #84 shows the Swap Rule is the condition to stay on the line; check it on the three-layer window. Leads to: the last exception of the law explained. Outside: T. J. McLarnan … Rep. Prog. Phys. 45 (1982) 587.
7. **Star and Star Halo of the classical solids**. Why near: the census scripts exist and #80 has the gains. Leads to: a full map of known and new stars and halos. Outside: *none found yet: a search target*.
8. **Galois twins of every golden cut**. Why near: #88 and #89 give the depths in closed form; only their minimal polynomials are needed, as in #90 — SETTLED 4 Nov 2026 (studies/2026-11-04-golden-twins): only the cube's two golden cuts are twins; the dodecahedron's greater cut (x = 1/3) is its own twin, its lesser cut's twin is the φ^3-cut dodecahedron (not a golden-cycle station); T, O, I twins over-cut. Leads to: a Double Vision column for every golden pair. Outside: M. Freedman et al. … Phys. Rev. B 85 (2012) 045414; K. A. Adiprasito … Invent. Math. 199 (2015) 581.
9. **Dehn counts of the rhombic dodecahedron, cuboctahedron, triacontahedron and icosidodecahedron pyramids**. Why near: #44's counting carries over — ANSWERED 4 Nov 2026: among Platonic and Catalan solids only the cube, RD and RT have Dehn-zero centre pyramids (their dihedrals 90°, 120°, 144° are the only rational ones); cube and RD pyramids tile, RT open (overnight search). Leads to: which pyramids can fill space beyond Plato. Outside: J.-P. Sydler, Comment. Math. Helv. 40 (1965) 43.
10. **The TRD as the rhombic dodecahedron's h partner**. Why near: the Rule of Halves (#57) and #93's Theorem C give the template — STUDIED 4 Nov 2026 (studies/2026-11-04-trd, note The Turned Twin, read ok): RD/TRD = ½ polar of CO/anticuboctahedron; TRD = RD half-turned 60°; equal pyramid volumes; DICTO's turned node = the twisted ball; ring cells drawn. Leads to: the TRD DICTOnode in a paper; the rhombic dodecahedron on every stacking. Outside: W. G. Burgers, Physica 1 (1934) 561.
11. **#53 Observation 1 (diagonal swap)**. Why near: switch points are walls between regular subdivisions, located by Edelsbrunner–Shah's lifted determinant — PROVED 8 Nov 2026 (studies/2026-11-08-switch-theorem: Theorems A–C; Obs 1 and 4 proved,

quadrilateral version disproved, Obs 3 partial, Obs 2 open; placement awaits DICTO) — Observation 2 PROVED 9 Nov 2026 (studies/2026-11-09-cscl-zincblende, Theorem E). Leads to: a general switch theorem for the bonds. Outside: F. Aurenhammer … Algorithmica 15 (1996) 223.

12. **The golden split in the other AXE networks (#48)**. Why near: the method and uniqueness theorem exist. Leads to: new convex pairs. Outside: *none found yet: a search target.*
13. **Archimedean stretches (#49)**. Why near: the Polyhedral Stretching Theorem covers every equal-edged solid; listing and prior-art check remain. Leads to: new equal-edged solids. Outside: *none found yet: a search target.*
14. **The loose step O ⊂ rTI (#35)**. Why near: a golden-cut station of #91 may fit it (unsure). Leads to: the Vajrayana cycle complete. Outside: P. P. Dechant … J. Phys. A 45 (2012) 285202.
15. **Joining the fill thread to the cuts and the sweep**. Why near: #89 §7 and #93 are already fillings by the sweep's pieces. Leads to: #47 questions 1 and 3; the isolated fill thread tied in. Outside: J. T. B. Overvelde et al. … Nature 541 (2017) 347; J. B. Berger … Nature 543 (2017) 533.

5.3 Open points left by the papers

Of about 60 open points stated in the papers and studies, 9 are closed by later papers (among them #77 and #44 face to face by #92, #38 by #38a and #39, #48 by #51, #56 by #57). The rest are listed here.

Paper	Open point	Status
#92, #93	reversed neighbour of the T and C6 pyramids; undecided rows of #92 Table 2; tilings not face to face (#93: 7 edge and 3 face combinations); TRD centre pyramids	open
#38a	do its shells fill by one pattern	CLOSED 9 Nov 2026 (studies/2026-11-09-harmonic-gaps)
#48	golden split in the other AXE networks; printed keyed sets	open
#35	closing solid for the loose step O ⊂ rTI; the 9/φ⁴ coincidence; a physical build	open
#36	golden fill G′ of the bcc gap	paused
#36	golden Bull-horns adaptor; rhombic division of the 72-piece gap; Cleo-Ico adapters; oblique layers at the seams	open
#30	spine = thinned cut-and-project chain; seams as model sets; off-plane core rings; other solids with five inscribed cubes	open
#40	space DBT and OTT leave on mixed stackings; OG on aperiodic stackings with diffraction	open
#41	third level of Hexas; physical puzzle	queued
#42	what fills between the great star and the Sunstar	CLOSED 9 Nov 2026 (studies/2026-11-09-harmonic-gaps)
#43	Conjectures 1 and 2; shell 5 exact; the full flip	open
#43	shell ratios against Tsai clusters and virus capsids	queued
#45	Conjecture 1 (largest width)	open
#47	smallest kit; face to face; periodic in three directions (C + T + O without connectors ANSWERED 6 Nov: yes, never face to face)	open
#49	Archimedean stretches and caps; prior art for the five-fold stretch	open
#50	catalogue of perfect points; which dissection fillers (Hill, Sommerville, Goldberg) the moves reach	open
#51	one solid per bond that fills alone; printed Samson and Delilah	open
#52	scale anomaly; web of the keyed bond	open
#53	Observations 1–4; exact walks of three-kind networks; a common reason for the switch points	open
#54	a shorter run as a classical gnomon; capped cubes at other sizes	open
#55	weight walks of σ, Z, C15 and other Frank–Kasper phases; 4D two-kind walks	partly queued

Paper	Open point	Status
#56, #57	self-keyed Blockhead; three-kind walk; keyed Chucky and Fang	open
#61	Swap Rule for s < 1/4	open
#66, #67	Hexagonality Law for the Laves big atom on every stacking (exact to 6 layers)	open
#74	Loop Equality for all lengths (to length 34)	open
#76	exact atlas in Q(√5); proof of part (3)	open
#78	why the prime 89; matching the dodecahedra to the Red Queen's nine layers	open
#80	proof that Archimedean chains close in four steps; omnitruncates; the quasi-regular formula	open
#80, #81	exact edges of the open windows (cube s 49/80–59/80; dodecahedron 87/80–99/80)	open
#86	proof over whole stretches (exact at 402 weights)	open
#87	other centres for solids of low symmetry	open
#87	cardboard sheets for the other solids	queued
#91	Lemma 6 in general; off-base prism sections; exact stations of the tilted slice; TRD outside the cubic sweep; the factor 2 between 342 and 171	open
#89, #91	sections of the ten Schläfli–Hess star polytopes	queued
#90	structure over Q(√2, √5) beyond natural projections	open
#8	database search for the 1b + 12j point set	open
#62	six all-triangle 600-cell slices (28, 30, 32, 42, 48, 72 vertices); equilateral forms of five patterns	open

5.4 Weak joins

- Concepts in one paper only: Window Law (#84), tower move (#86), knitting slice (#83), Vacuus (#87), field obstruction and cut equals shadow (#90), polyhedral stretching (#49), lattice moves (#50).
- Cross-links still only suggestive: #48 ↔ #55 (one pyritohedral family at t = φ, φ², 2); #89/#91 ↔ #51/#53/#57 (the tesseract slice meets the fluorite lane only at the octet); #43 Conjecture 2 ↔ #80 (constant 3); #12 ↔ #38a.
- Threads that share no paper and no concept: fill with kepler, cycles, og, bonds, stack, queens, cuts and sweep; kepler with queens. The fill thread is the isolated one, partly a matter of tagging (#89 §7 and #93 are fillings).

5.5 From elsewhere: what other fields hold

Results from other fields, under other names, that bear on the threads and are not cited in any paper. Kind: **credit**, earlier or parallel work to cite; **test**, a check to run in the engine; **lead**, a direction to follow; **anchor**, the same mathematics in a real object or measurement. The strongest: the big bang billiard is the one triangle at p = ∞, so the ladder runs sphere (p = 3, 4, 5), plane (6), hyperbolic (7 and up), big bang (∞); the numbers 2 cos(π/p) are measured particle and anyon dimensions; and diamond growers' shape dial is α = 3/μ, our core turn.

Field	Source	What it holds	Touches	Kind	Library	Checked	Status
Physics: the big bang	T. Damour, M. Henneaux, H. Nicolai, Cosmological billiards, Class. Quantum Grav. 20 (2003) R145, arXiv:hep-th/0212256	Near the big bang singularity gravity becomes a ball bouncing in a mirror chamber of the Kac–Moody group AE3; its angles are 90°, 60° and 0°, the (2, 3, ∞) triangle (checked from the Cartan matrix).	one triangle (§4); all threads	lead	to fetch	abstract; angles checked	to read
Physics: the big bang	A. J. Feingold, A. Kleinschmidt, H. Nicolai, Hyperbolic Weyl groups and the four normed division algebras, J. Algebra 322 (2009) 1295, arXiv:0805.3018	The billiard groups of ranks 3, 4, 6, 10 are modular groups over the integers, Gaussian integers, Hurwitz quaternions (the 24-cell) and octonions.	queens; one triangle	lead	to fetch	abstract	to read

Field	Source	What it holds	Touches	Kind	Library	Checked	Status
Physics: particles	A. B. Zamolodchikov 1989; R. Coldea et al., Quantum criticality in an Ising chain: experimental evidence for emergent E8 symmetry, Science 327 (2010) 177	In a critical magnetic chain the first two particle masses are in the ratio 2 cos(π/5) = φ, measured in CoNb2O6.	one triangle (p = 5)	anchor	to fetch	abstract	to read
Physics: quantum matter	A. Feiguin et al., Interacting anyons in topological quantum liquids: the golden chain, Phys. Rev. Lett. 98 (2007) 160409; V. F. R. Jones, Index for subfactors, Invent. Math. 72 (1983) 1	Anyon dimensions 2 cos(π/(k + 2)) are 1, √2, φ, √3 for k = 1 … 4, our p = 3 … 6; Jones's indices are their squares.	one triangle	anchor	to fetch	abstract; standard	to read
Physics: quantum matter	M. Freedman et al., Galois conjugates of topological phases, Phys. Rev. B 85 (2012) 045414	The sign swap √5 ↦ −√5 sends the Fibonacci theory (φ) to a non-unitary Yang–Lee partner (−1/φ): Double Vision in physics, with only one twin physical.	cuts (Galois twins); Double Vision	lead	to fetch	abstract	to read
Physics: antimatter	Charge conjugation as −w0 (J. E. Humphreys, Reflection Groups and Coxeter Groups, 1990)	For the cube, octahedron, dodecahedron and icosahedron symmetries the antiparticle map is trivial; only the tetrahedron's differs: its point mirror −T, which is also its polar (the 4 and 4̄ of SU(4)).	kepler (stella octangula); sweep	lead	to fetch	standard; geometry checked	to read
Physics: particles	P. P. Dechant, The birth of E8 out of the spinors of the icosahedron, Proc. R. Soc. A 472 (2016) 20150504; Dechant, Adv. Appl. Clifford Algebras 23 (2013) 301	The 3D symmetries induce the 4D ones: tetrahedral → 24-cell, cube–octahedron → F4, icosahedral → 600-cell and 120-cell, and E8 arises from the icosahedron.	queens	credit	to fetch	abstract	to read
Physics: particles	P. P. Dechant, C. Bœhm, R. Twarock, Novel Kac–Moody-type affine extensions of non-crystallographic Coxeter groups, J. Phys. A 45 (2012) 285202	Golden affine extensions of the icosahedral group with translations classified by Fibonacci recursions.	cycles (φ³ step: test)	test	to fetch	abstract	to read
Physics: holography	F. Pastawski et al., JHEP 06 (2015) 149; L. Boyle, M. Dickens, F. Flicker, Conformal quasicrystals and holography, Phys. Rev. X 10 (2020) 011009	Regular hyperbolic tilings carry quantum codes whose boundary is a quasicrystal: periodic inside, aperiodic on the edge.	one triangle (p ≥ 7); periodic–aperiodic aim	lead	to fetch	abstract	to read
Physics: particles	L. Boyle, P. J. Steinhardt, Coxeter pairs, Ammann patterns and Penrose-like tilings, Phys. Rev. B 106 (2022) 144113	Pairs H3/D6 and H4/E8 give Penrose-like tilings in 3D and 4D from six- and eight-dimensional lattices.	sweep (6D sections); queens	test	to fetch	abstract	to read
Crystal growth	C. Wild et al., Oriented CVD diamond films: twin formation, structure and morphology, Diamond Relat. Mater. 3 (1994) 373; K. Hermann, J. Phys.: Condens. Matter 36 (2024) 045303	Diamond growers set crystal shape by α = √3 v100/v111 from cube (1) through cuboctahedron (1.5) to octahedron (3). For our sweep α = 3/μ exactly, so the core turn μ ↔ 3/μ is the switch between the two dials. Hermann maps the wider {100}{110}{111} shape diagram.	sweep	anchor	yes	abstract; α = 3/μ checked; Hermann in library	to read
Materials	J. B. Berger, H. N. G. Wadley, R. M. McMeeking, Mechanical metamaterials at the theoretical limit of isotropic elastic stiffness, Nature 543 (2017) 533	Walls on the cube and octet planes make the stiffest possible lattice.	pyr (Vacuus SHEETS); fill	credit	to fetch	secondary	to read
Materials	A. V. Dyskin et al., Topological interlocking of platonic solids, Phil. Mag. Lett. 83 (2003) 197	Layers of each Platonic solid can lock so that no block can be lifted out.	pyr (Vacuus); kepler; DICTOnodes	test	to fetch	abstract	tested
Materials	J. T. B. Overvelde et al., Rational design of reconfigurable prismatic architected materials, Nature 541 (2017) 347	Space-filling tessellations extruded into folding materials.	fill (#47 kits)	lead	to fetch	abstract	to read

Field	Source	What it holds	Touches	Kind	Library	Checked	Status
Tilings	M. Goldberg, Three infinite families of tetrahedral space-fillers, J. Combin. Theory A 16 (1974) 348; A. L. Edmonds, Sommerville's missing tetrahedra, Discrete Comput. Geom. 37 (2007)	Families of space-filling tetrahedra; the face-to-face orientation-preserving ones are exactly Sommerville's four.	pyr (face-to-face law)	test	to fetch	secondary	to read
Scissors congruence	J.-P. Sydler, Comment. Math. Helv. 40 (1965) 43	Equal volume and equal Dehn invariant imply cut-and-glue equality: the converse of Dehn.	fill (#50 exact vs cut-and-glue)	credit	to fetch	secondary	to read
Stacking statistics	T. J. McLarnan, Z. Kristallogr. 155 (1981) 269; E. Estevez-Rams et al., Acta Cryst. A 73 (2017); M. E. Fisher, W. Selke, Phys. Rev. Lett. 44 (1980) 1502; P. Bak, Rep. Prog. Phys. 45 (1982) 587	Counts of stackings by period; block statistics of faulted stackings; devil's staircases with irrational (Fibonacci-like) points.	stack (Window Law, Swap Rule, Fibonacci stacking)	credit	to fetch	abstract / secondary	to read
Four dimensions	T. Denney et al., The geometry of H4 polytopes, Adv. Geom. 20 (2020) 433; H. S. M. Coxeter, The simplicial helix, Canad. Math. Bull. 28 (1985) 385	The 25 24-cells in the 600-cell and its decagons; the tetrahelix rings (Hopf fibration).	queens; cycles	test	to fetch	abstract	tested
Polytope theory	K. A. Adiprasito, G. M. Ziegler, Many projectively unique polytopes, Invent. Math. 199 (2015) 581	Polytopes realisable over a given number field and no smaller one.	cuts (#90 field obstruction)	credit	to fetch	abstract	to read
Convex geometry	H. Iriyeh, M. Shibata, Duke Math. J. 169 (2020) 1077	Mahler's volume product $\|K\|\cdot\|K^\circ\|$ is least exactly at the cube and the octahedron in 3D. Tested 4 Nov 2026 (study 2026-11-04-mahler-sweep): along the sweep the product is 32/3 at both ends and largest at the root $\mu \approx 1.6316$ of $2\mu^3 - 9\mu + 6 = 0$, just past Icosaxes, not golden.	sweep (test along μ)	test	to fetch	abstract	tested
Chemistry	M. J. Moses, J. C. Fettinger, B. W. Eichhorn, Science 300 (2003) 778	The ion [As@Ni12@As20]3−: an atom in an icosahedron in a dodecahedron, Kepler's nest as a molecule.	kepler; cycles	anchor	to fetch	secondary	to read
Chemistry	W. I. F. David et al., Nature 353 (1991) 147; O. Zhou et al., Nature 351 (1991) 462; A. Y. Ganin et al., Nature Materials 7 (2008) 367	C60 (20 hexagons) packs fcc with hexagons on the octet's octahedra; doped forms pack bcc and A15.	og (#30, #36, #37); bonds (#55)	anchor	to fetch	abstract	to read
Nanoparticles	S. Ino, J. Phys. Soc. Japan 27 (1969) 941	Five fcc wedges round a five-fold axis leave a 7.35° gap.	og (#30)	credit	to fetch	abstract	to read
Quasicrystals	T. Ogawa, J. Phys. Soc. Japan 54 (1985) 3205	A 3D Penrose inflation with step φ^3 in length.	cycles (φ^3 step)	credit	to fetch	abstract	to read
Quasicrystals	L. Danzer, Discrete Math. 76 (1989) 1; A. L. Mackay, Physica A 114 (1982) 609; A. Janner, Acta Cryst. A 62 (2006)	The ABCK golden tetrahedra under inflation; the 3D Penrose pattern from interpenetrating triacontahedra; nested golden polyhedra in virus capsids.	cycles (#24 check against Mackay)	test	to fetch	abstract / secondary	to read
Computational geometry	F. Aurenhammer, F. Hoffmann, B. Aronov, Algorithmica 20 (1998) 61; H. Edelsbrunner, N. R. Shah, Algorithmica 15 (1996) 223	Weights giving prescribed cell volumes always exist (optimal transport); regular triangulations change by flips.	bonds (#51 equal volume; #52, #53 stops)	credit	to fetch	abstract	to read
Metallurgy	W. G. Burgers, Physica 1 (1934) 561	The bcc → hcp path, companion of Bain's fcc ↔ bcc.	bonds (#56, #57)	credit	to fetch	secondary	to read
Art and stellation	Escher's solid (M. C. Escher, Waterfall, 1961; G. W. Hart, The polyhedra of M. C. Escher)	The first stellation of the rhombic dodecahedron fills space alone.	kepler (DICTO Jewel fills)	test	to fetch	secondary	to read

Field	Source	What it holds	Touches	Kind	Library	Checked	Status
Islamic art: Persian tilings	P. J. Lu, P. J. Steinhardt, Decagonal and quasi-crystalline tilings in medieval Islamic architecture, Science 315 (2007) 1106, doi:10.1126/science.1135491	Five decorated girih tiles built the strapwork patterns; the Darb-i Imam shrine in Isfahan (1453) carries a nearly perfect Penrose-type quasicrystalline pattern, with a self-similar subdivision.	kepler and cycles (decagonal, golden inflation); periodic–aperiodic aim	anchor	to fetch	abstract; later critiques of the Darb-i Imam mapping exist	to read
Islamic architecture: muqarnas	Aboufadil et al., Acta Cryst. A 75 (2019) conference abstract; Dinçer, Korumaz, Yazar, Anatolian Muqarnas (ACM, 2024); Y. Turan, Symmetry: Culture and Science (2025); M. Dold-Samplonius, Centaurus 35 (1992), al-Kashi's measure of the muqarnas	The muqarnas is a layered tiling of space by a few concave units whose plan is a plane tiling; Anatolian Seljuk plans use square–rhombus (Ammann–Beenker type) tessellations.	pyr (layered units over a plane tiling); stack (layers); fill	lead	to fetch	abstracts / secondary	to read
Weaving and textiles	D. K. Washburn, D. W. Crowe, Symmetries of Culture (1988); Washburn, Crowe (eds.), Symmetry Comes of Age (2004), with B. Grünbaum on Peruvian textiles	The plane symmetry groups as a working method for woven and knotted patterns; the knot grid of a carpet is a square lattice, the kaleidoscopes p = 4 and the flat p = 3, 6 in cloth.	one triangle (plane groups); 2D kaleidoscope worlds	lead	to fetch	secondary; no source on carpet knot grids found	to read
Music theory	D. Tymoczko, The geometry of musical chords, Science 313 (2006) 72; the Tonnetz (Euler 1739, Riemann)	The Tonnetz is the triangular lattice, our flat triangle p = 6; chord spaces are orbifolds, spaces folded by mirrors, i.e. kaleidoscopes.	one triangle (p = 6); stack (the triangular layer)	lead	to fetch	secondary	to read
Waves and resonance	A string of p − 1 equal beads (the path graph, which is also the Coxeter diagram A(p−1)): its highest vibration has frequency number 2 cos(π/p)	So the mirror numbers 1, √2, φ, √3 of the one triangle (p = 3, 4, 5, 6) are the top resonances of strings of 2, 3, 4, 5 beads.	one triangle	anchor	to fetch	checked (eigenvalues, exact formula 2 cos(kπ/p))	to read
Waves: spherical harmonics	“Spherical harmonics and the icosahedron”, arXiv:0706.0088; P. J. Steinhardt, D. R. Nelson, M. Ronchetti, Bond-orientational order in liquids and glasses, Phys. Rev. B 28 (1983) 784	Icosahedral symmetry allows spherical harmonics only of degrees 6p + 10q and 6p + 10q + 15 (first at 6); the degree-6 measure Q6 detects icosahedral order in liquids and glasses. Tested 4 November 2026 (study 2026-11-04-harmonic-fingerprint): on the cube–octahedron sweep the cubic part Q4 vanishes only at Icosaxes, μ = φ (exact: $a^4 - 3a^2 + 1 = 0$), whose 24 corners are two icosahedra.	cycles, og, queens (harmonic fingerprint of the nests)	test	to fetch	abstract; Steinhardt et al. standard	tested

5.6 Waiting for their volume

Results found since the last version, with their study folder and the volume or version they go into (released as soon as DICTO has read them).

Result	Study	Goes to
Reversed neighbour impossible for T and C6: whole tetrahedra, whole cubes in every tiling	2026-11-04-reversed-neighbour	#92 v3 and #93 v2 (Centre Pyramids volume)
#92 Table 2 complete: the three undecided edge rows have an odd number of T base wedges, impossible once T pyramids form whole tetrahedra (#92 open point 2 closed)	2026-11-04-reversed-neighbour	#92 v3
RD pyramids not face to face: planes x + y = k made of faces in a Theorem C tiling, so half-spaces slide freely: uncountably many, no finite list (reversed neighbours occur for R12: they are Theorem C)	2026-11-04-reversed-neighbour	#93 v2
Admiral Cap network: frustrated (at most 8 of 12 shared edges round an octet corner); ground states alternate along one axis with a free flip per layer across another; twin-boundary node of two icosahedra	2026-11-04-network-fingerprint	Space-Fillers and Lattice Moves volume (#47 world)
Galois twins of the golden cuts (cube only; dodecahedron self-twin and φ^3 cut); φ^3 cut not a golden-cycle station	2026-11-04-golden-twins	The Golden Ladder
Golden Ladder chapter 1, One Triangle, Three Numbers (read ok)	2026-11-04-golden-ladder-ch1	The Golden Ladder v1

Result	Study	Goes to
The Turned Twin: TRD, turned node, ring cells (read ok)	2026-11-04-trd, 2026-11-04-ring-rd	HexOct volume
Harmonic fingerprint: CO–RD sweep, tetrahedron sweep, network reading of c/h; #94 Figure 1 labels fixed	2026-11-04-harmonic-fingerprint, 2026-11-04-co-rd-sweep	#94 v2
PET-ring RD cells: tetrapod of 4 RDs, build sheet for 1/4/6/14 cells, RD and TRD ring cells	2026-11-04-ring-rd	Quanta (nodes) / HexOct volume
Consolidation: one living volume per thread	2026-11-04-consolidation/PLAN.md	all volumes
Octahedron pyramids: reversed neighbours DO occur (a cube of the octet's cube lattice cut the other way, #92 Prop. 8). Conjecture: every O pyramid lies in a whole octahedron or is a corner of a whole cube with its tetrahedron. Needs an argument controlling sliding edges (as #92 Cor. 9); not a queue job yet	2026-11-04-reversed-neighbour	open: next session
Mahler volume product on the cube–octahedron sweep: exact formula; 32/3 at the ends; maximum at $2\mu^3 - 9\mu + 6 = 0$ ($\mu \approx 1.6312$), near but not at Icosaxes	2026-11-04-mahler-sweep	#94 v2 or The Golden Ladder (sweep)
Parallel outside tests on the sweep (DICTO's idea): the isoperimetric maximum is exactly $\mu = \sqrt{3}$, the fixed point of the core turn $\mu \leftrightarrow 3/\mu$ (where diamond's $\alpha = \mu$); harmonic Q4 = 0 exactly at Icosaxes φ; Mahler maximum near φ but cubic. Table of special points	2026-11-04-mahler-sweep	#94 v2 / The Golden Ladder (sweep)
Harmonic fingerprint in 4D: calibrated on the regular polytopes (600-cell silent through degree 11, a spherical 11-design); #91's 4D sweep: Q4 = 0 only at the 24-cell ($\mu = 2$, fixed point of $\mu \leftrightarrow 4/\mu$), Q6 = 0 exactly at $\sqrt{2}$ and $1 + \sqrt{3}$; All-Golden stations not seen (golden content in the cells)	2026-11-04-harmonic-4d	#94 v2 / Alicia (Queens) volume
600-cell slices fingerprinted: corner-first slices always silent in Q4, Q8, and silent in Q6 at $\lvert h\rvert \approx 0.128, 0.396, 0.638$; cell-first slices silent in Q4 at four heights, in Q8 at four others, never together (no hidden icosahedral slice); exact heights open	2026-11-04-harmonic-4d	Alicia (Queens) volume
Parallel tests on the CO–RD sweep: Mahler 40/3 at both ends, maximum at $12\mu^3 - 36\mu^2 + 43 = 0$ ($\mu \approx 1.600$), roundness maximum ≈ 1.523: no named station; the exact $\sqrt{3}$ coincidence is particular to the cube–octahedron pair	2026-11-04-mahler-sweep	The Golden Ladder (sweeps)
Harmonic test on the Delian walk (#51): the octahedral piece O(t) is cubic-silent exactly at t = 1/9, but never together with Q8; no hidden icosahedral station on this walk	2026-11-04-harmonic-fingerprint	#94 v2 / DICTONIC Bonds volume
Foam test on the CsCl pairs: wall area per cell is symmetric under $\mu \leftrightarrow 3 - \mu$, largest exactly at Kelvin's cell ($\mu = 3/2$, the only equal-volume pair), smallest at the cuboctahedron + octahedron end ($(3 + 2\sqrt{3})/4$)	2026-11-04-mahler-sweep	#93 v2 / Space-Fillers volume
OG family fingerprinted: icosahedral (Q4 = Q8 = 0); W6 sign splits it into corner-type TQHD, TQP, OG, KRE and face-type TQH, TPH, PHE, TPP, DPHP, changing in pairs along the chain; KRE the quietest	2026-11-04-nest-fingerprint	OG volume
Red Queen (120-cell): an 11-design like the White Queen. Path between the Queens: degrees 1–11 silent throughout; on the truncated-600-cell stretch (1440 exact corners) Q12 = 0 at $\mu \approx 0.92417$, making those 1440 points a spherical 19-design (the classical W(H4)-orbit 19-design, Goethals–Seidel 1981, Bannai survey — credited; ours: it is a station of the Queens path); Q20 = 0 near $\mu \approx 0.954$. Exact μ to compute	2026-11-04-harmonic-4d	Alicia (Queens) volume
Fibonacci stacking spectrum: the c/h word is pure point, every harmonic at q/φ^2 with exact amplitude $\lvert\sin(\pi q/\varphi^2)\rvert/(\pi q)$; the crystal it builds (A/B/C positions) has fading, non-Bragg peaks at l = 1/2 and l = cos 72° — probably singular continuous, like Thue–Morse	2026-11-04-fibonacci-spectrum	HexOct volume (stackings)
Anticuboctahedra (J27) and octahedra fill space on hcp, and with cuboctahedra on mixed stackings (ABCB, ABCACB): the twisted twin of the rectified cubic honeycomb; centres 1/4 of the balls (integer program on periodic boxes; prior art to check)	2026-11-04-cuboctahedron-pyramids	#92 v3 (Theorem 10 sequel) / HexOct volume
The cube–octahedron sweep is the family of weight shapes of SU(4) ≅ SO(6) (self-conjugate multiplets, $\mu = (2a + b)/(a + b)$, octahedron 6 … Kelvin 64 … cuboctahedron 15) and SO(7) ($\mu = (4b + 3c)/(2b + c)$, cuboctahedron 21 … cube 8): golden ladder of Fibonacci multiplets → Icosaxes and Unborn Axes; silver ladder of Pell multiplets (112, 4928, 905520, …) → the Archimedean truncated cube at $1 + \sqrt{2}$; Golden Cut Cubes quartic, no simple ladder	2026-11-04-su4-weights	#94 v2 / The Golden Ladder (sweep)
Tuning: the first half of the sweep is one octave (octahedron = unison, cuboctahedron = octave); every just interval is a self-conjugate SU(4) multiplet (fourth 175, fifth 64 = Kelvin, major sixth 300 …); the cubic tone is silent exactly at the golden interval (833 cents, Icosaxes), between the sixths. App idea: tuning instrument	2026-11-04-su4-weights	The Golden Ladder / app
#91's 4D sweep is the family of SO(9) weight shapes: 16-cell = vector 9, 24-cell = adjoint 36, rectified tesseract = 84, tesseract = spinor 16; DICTO's All-Golden Polytopes ($\mu = \varphi^2$, $2 + 1/\varphi^2$) are the limits of Fibonacci multiplets (0, F_n, F_{n+1}, 0): dims 1650, 27027, 1527552, …	2026-11-04-su4-weights	Alicia (Queens) / The Golden Ladder
Golden Ladder chapter 2, Silence Is Golden (DICTO's title; subtitle The Ladder of Multiplets), read ok: sweeps as weight shapes of SU(4), SO(7), SO(9); Fibonacci and Pell ladders; one octave	papers/golden-ladder-ch2	The Golden Ladder v1 (with chapter 1), weekly batch
Dehn test (gap 9): exactly three isohedral solids have Dehn-zero centre pyramids — cube, RD, RT; the RT pyramids are the open case (search alive, overnight run)	2026-11-04-rt-pyramids	#93 v2 / Centre Pyramids volume

Result	Study	Goes to
Dehn test on the sweeps: cube–octahedron sweep zero only at Kelvin and the cube (its lone tilers); CO–RD sweep zero only at μ = 5/3, the B3 permutohedron (±1, ±2, ±3) (ρ of Sp(6)) — which the edge-angle count shows cannot tile alone (135° edges cannot close): Dehn zero is not sufficient	2026-11-04-rt-pyramids	Space-Fillers volume
Hopf rings of the White Queen: 12 decagons over the icosahedron, 20 hexagons over the dodecahedron, 30 squares over the icosidodecahedron (classical; the golden cycle's first stations read off the Queen)	2026-11-04-hopf-interlock	Alicia (Queens) volume / app view
Interlocking test: in an octet (111) slab every octahedron is locked, every tetrahedron free (slides out obliquely); lead: Vacuus, HexOct and DICTO's builds	2026-11-04-hopf-interlock	Space-Fillers volume
Interlocking (slides and twists, exact contact polygons): every Admiral Cap and icosahedron in the #47 slab is locked — the slab holds itself by shape alone (Dyskin's topological interlocking) apart from 4 surface tetrahedron pyramids; calibration Dyskin tetrahedra locked	2026-11-04-hopf-interlock	Space-Fillers volume (#47 chapter) / builds
HexOct tiles interlock: a single layer of c-tiles or of h-tiles (#75, λ = 1/10) is locked against slides and twists (all inner tiles), with positive (Dyskin) and negative (upright cubes free) controls — HexOct tiles are interlocking tiles	2026-11-04-hopf-interlock	HexOct volume
Tetrahedron sweep tests: roundness max at the octahedron; Mahler exact, max at (11−√33)/4; Dehn zero only at μ = 2 (T(4) minus corners), which tiles space alone (octet exact cover, verified with real polytopes) — a KNOWN tiling (Damasceno–Engel–Glotzer 2012, β-tin; Gabbrielli–Jiao–Torquato 2012 Fig. 15); new link: Dehn zero = octet 2:1 ratio at n = 4	2026-11-04-tetra-sweep	Centre Pyramids / Space-Fillers volume (Dehn screen calibration) and #94 v2
Roundness on the cube ∩ μ·cross sweep in n dimensions: maximum exactly at μ = √n (n = 2..12 exact; octagon, √3, 24-cell) — explained by Lindelöf's theorem (facets tangent to one ball). CORRECTION: the earlier '√3 exact coincidence' is this classical theorem; #94 v2 wording to change	2026-11-04-tetra-sweep	#94 v2 / The Golden Ladder (sweep)
General Dehn screen (angle relations by PSLQ, calibrated 11/11): all nine OG-family solids have nonzero Dehn invariant — none tiles space alone or is scissors-congruent to a cube (not automatic: the RT is icosahedral with Dehn zero)	2026-11-04-dehn-screen	OG volume
Dehn screen on the HexOct tiles: whole tiles zero (calibration); c-tile pieces nonzero; h-tile pieces nonzero below λ = 5/24 and ZERO from 5/24 on (octahedral piece 10 corners until it vanishes at 3/8) — the screen finds the known 5/24 stretch independently; lead: do h-pieces tile alone there?; for λ ≥ 3/8 the h-tile is two mirror copies of ONE fixed 10-corner solid (hexagon top, apex on a ball, dihedrals 90/120/135, volume half a ball cell), which therefore tiles space alone — identity/prior art to check	2026-11-04-dehn-screen	HexOct volume
Dehn screen on the Admiral Cap network: Q′, I and the A cap each have nonzero Dehn invariant — no kind fills space alone; only the mixture can	2026-11-04-dehn-screen	Space-Fillers volume
Dehn screen on the Delian walk (#51): O(t) nonzero for every t > 0 (single irrational angle 144.74°, no cancellation), zero only at the RD — agrees with #93 (walk rigid except at the octet)	2026-11-04-dehn-screen	DICTONIC Bonds volume / #93 v2
Golden Dehn-zero truncated tetrahedra: seven cut vectors in powers of 1/φ summing to 1 (exact in Z[φ], to φ^{-8}); none harmonically silent (Q3–Q8); Fibonacci approximants under tiling test (census-golden, running)	2026-11-04-tetra-sweep	Centre Pyramids / Space-Fillers volume; The Golden Ladder
Aperiodic-monotile clue checklist (DICTO 4 Nov): necessary Dehn zero, angle count, density; signals forced irrational frequency, forced 5-fold edges, golden self-similarity, search signature; candidates RT centre pyramid (convex: Tsiokos 2026 open Q3) and golden truncated tetrahedra	open-work 11-04	watch list (no claim)
Every Dehn-zero truncated tetrahedron tested (6 of 6: (1,1,0,0), (1,1,1,0), (2,1,0,0), (2,2,0,0), (2,1,1,0), (1,1,1,1)) tiles space, always piece + point inverse, 2 per period (SAT exact cover on the octet). Conjecture: Dehn zero ⇔ tiles for the whole family; prior art: only the uniform cut (1/4) is in the literature checked	2026-11-04-tetra-sweep	Space-Fillers / Centre Pyramids volume
Outside lead (CMB): the White Queen's 120 icosians are the Poincaré dodecahedral space's group; from our corners the invariant-harmonic counts match Aurich–Lustig–Steiner at every degree k ≤ 120; our 11-design result = the PDS's missing low modes (classical, cross-checked; no cosmology claim)	2026-11-04-pds-modes	Alicia (Queens) volume / #94 v2 (outside link)
Narcissus alone stacks any straight/turned word; 17 keyed Narcissus (14 Kari tile pieces + 3 relays) force a non-periodic stacking word (density proof; SAT on real geometry agrees)	2026-11-05-one-piece-any-stacking	Narcissus and Covenant (HexOct chapter) or its own short paper — to ask
Narcissus is Goldberg's space-filler type 10-XII (1982), formula 1036-64; half the RD is a different type	2026-11-04-halves / prior art	published in #98, Narcissus and Covenant (HexOct chapter)

Result	Study	Goes to
2026-11-06 gap 2 CHECKED (corrected same day): the near-miss 'rectified truncated icosahedron' (60{3}+12{5}+20{6}) is listed in Wikipedia's near-miss table and on McCooey's page. McCooey's form has REGULAR hexagons, regular pentagons and isosceles triangles, 60 short + 120 long edges (long/short = (sqrt15-sqrt3)/2): that is our cut OG on one sphere (#95 Theorem 1, already marked rediscovered). Our ALL-EQUAL-EDGE form (equilateral triangles, regular pentagons, equilateral hexagons 124.72/115.28) is NOT given in either source: arrangement known, this exact solid not found in the sources checked (Wikipedia 'Near-miss Johnson solid', dmccooey.com RectifiedTruncatedIcosahedron). Home: #95 v2 table row 'All-equal-edge version' -> 'arrangement known; this form not found (Wikipedia, McCooey)'.		
2026-11-06 Narcissus vs Goldberg 10-XII: Goldberg 1982 (p.43 caption, read from the free UPC scan) defines 10-XII as 'join of two opposite-hand 7-XXIX at lateral faces'. Narcissus cut on a mirror plane = two mirror-image heptahedra, volume sqrt2/8 each (half.py, float). So Narcissus fits Goldberg's construction form; whether its half is in the 7-XXIX class needs Goldberg 1978 (Geom. Dedicata 7, 175-184; Springer paywall, no legitimate free copy found; UPCommons holds only the 1982 paper). Home: Narcissus and Covenant prior art.		
2026-11-06 Golden OGs (#91, published): same status as OG: face pattern (20 hexagons + 60 kites, 102 corners) known (Rossiter/McCooey); golden metric forms not found in sources checked (Wikipedia near-miss table, web search 'polyhedron 20 regular hexagons 60 kites 102 vertices').		
2026-11-06 Route (a) step 1 (count): one keyed Narcissus has <= 6 placements per site (3 without mirror copies); Kari x2/3 slabs need 10 tile states per down site, so one piece cannot carry one tile per site. Tile+relay PAIRS need 10 x 3 = 30 states: with mirror copies 36 >= 30 passes; without them it needs >= 4-site clusters. Next: a SAT search for one key pattern with a pair decoding. Home: #97 follow-up (one-piece paper).		
2026-11-06 PUBLISHED The Golden Ladder v2 (record 23990126; concept 10.5281/zenodo.23966554): whole book, 95 pp, map + ch1-2 + 3A #20, 3B #48, 4 #88, 5 #65, 6 #89, 7 #90, 8 #91, 9 E8 closing; ladder cover A by DICTO. Site shelf live 3289355. Open: ORCID line; Collected Works next weekly version lists it.		
YES: octet layers (close-packed planes) and cube layers are flat slabs; stacked they fill space with no connector (checked: 200 000 points, studies/2026-11-06-cto-no-connectors). Never face to face (only the cube has square faces). The slab stack is not 3-periodic; 3-periodic C+T+O open.		#47 version 3 (published)
Zenodo 24005568 (concept 24005567), 6 Nov 2026		site paper tile (deploy queue, ask)
Zenodo 24005562: Theorem 2 C+T+O with no connector, Figure 5		done
Zenodo 24005930 (concept 24005929), 6 Nov 2026		HexOct versions 8–11 (published); site tile done
O 1 > I sqrt2/phi^2 > D > C sqrt2/3 (= the octahedron's dual cube) > T 2/3 in every octahedron; T > TT 1/3 > O 1/2 > ... in every tetrahedron: both octet cells carry all five; 3-periodic. 12 inscriptions, 14 five-chains (18 with TT). studies/2026-11-06-five-in-octet		five-in-octet paper (theorem) / Space-Fillers
O-I Admiral Caps; I-D 12 Sheriffs + 30 Lone-stars (DICTO's names, #99); D-C Euclid roofs; C-T = 4 octahedron centre pyramids x2/3		five-in-octet paper
nested Dodeca-13 (turns aligned by mirroring), 600-cell corner (20 nested tetrahedra, gap 14.1%), UNITY nested (heart icosahedron 2/sqrt5 clear of caps); Dodeca-13 fits its own central dodecahedron at x1/3 (float). studies/2026-11-06-nest-supercells		five-in-octet paper / Dodeca-13 follow-up
cut-and-project rungs 1/1, 2/1, 3/2, 5/3 tiled by golden rhombohedra (10+6, 84+52, 356+220, 754+466; points = tiles); icosahedral seats by harmonics 1,1,6,16 per cell, spacing 4.5346, nests seated; diffraction ten-fold 0.916 -> 0.9974 (8/5) -> 0.9977 quasilattice. studies/2026-11-06-approximant-ladder		five-in-octet paper (ladder section)
rock-salt pieces on stackings: cubes / truncated cubes + regular octahedra edge 2(3/8-lambda) in cubic stretches; freedom in two deformed pieces at hexagonal layers; no regular tetrahedra. studies/2026-11-06-premise-challenge		HexOct / #47 next versions
pentagon-hexagon linked football layer at 79.19 deg; two orientations -> chains only. studies/2026-11-06-football-scutoids		scutoid paper
Zenodo 24012440 (concept 24012439), 6 Nov 2026; Sheriff and Lone-star named		site tile (deploy queue); book 'Platonic lattices' plan

Fields not yet consulted, named by the search: dislocation theory (each h ↔ c swap is a partial-dislocation glide); commensurate–incommensurate transitions; semi-discrete optimal transport (the weight walk); martensitic paths (Burgers, Kurdjumov–Sachs); topological interlocking and architected materials; realisation spaces of polytopes; hypersimplices (the octahedron is Δ(2, 4)); fulleride and cluster chemistry; multiply-twinned nanoparticles; hyperbolic tilings and holography.

Notes on this version

- Version 2.2 adds the papers #97–#103 to the list and to their threads, and an eleventh thread, Mandalas and yantras, opened by #103; it removes stale "draft" marks (#90 and #91 are published; the book corrections name the published versions; the waiting list names #98, #47 v3 and HexOct v8–11 as published), names the I−D nest pieces Sheriffs and Lone-stars, and adds DICTO's names for the unifying framework, the DICTO Hopscotch (Antipol, hop, step, jump, turn, emergence, dissolution).
- Version 2, 5 November 2026 (version 1, 4 November). Sources: every paper.src.html (else paper.html) in papers/, the paper cards of dictospheres.com, the DICTOnary atom cards and the cross-links audit of 4 November; papers to #96. Each new paper, read or test is one edit to concordance.json and a rebuild; each version is released as soon as DICTO has read it.
- Not yet covered: the books as entries of their own; the findings F1–F21 except where a paper, card or DICTOnary entry names them; the Kaleidohedra app Almanac (alm.c.*), whose concept ids these could feed later; every piece of every kit (only pieces with a name are listed).
- To check (threads): the seed list of the four-dimensional queens named #64, which is The Swap Rule in Crystals; it is replaced here by #62 and #63. There is no card numbered #82, #46 or #21.
- To check (dates): for #8–#38a the local texts carry only the date of their current version (20 October 2026, the licence change); #1–#20, #27 and #31a–#35b have no text of their own in papers/ (findings, book chapters or short records): their lines come from the site cards.
- Names: every name is DICTO's choice unless a paper says otherwise (DICTO, 4 November 2026). Fallen Humpty (D102): the DICTOnary gives #62, the name is also in #63. Golden Clasp (D107) first appears in a study, not a paper.
- Name clash: “golden tetrahedra” of #38a (Danzer, Mosseri–Sadoc) are not the Golden Cut Tetrahedra of #88/#89. The TI Star core (D105) is cut at $(10 - \sqrt{5})/19$ of each edge (#78 version 3); the DICTOnary is corrected (v1.3), the published Golden Cycle and Star Cores books at their next versions.
- To check (the triangle): the placement of each thread on page “The Singularity” is the compiler's reading of the papers, for DICTO to confirm.
- Part 5B: every outside find was checked against its abstract or a reliable record (column “checked”); none is yet read in full, all stand “to read”. Unrefereed preprints found by the search are left out. Library copies: only Hermann is held so far.

5.7 Overlaps across threads (4 November 2026)

Cross-thread overlaps found on 4 Nov 2026: the involution fixed points (√3, 3/2, 24-cell $\mu = 2$) are singled out by roundness, foam and 4D harmonics; φ appears as Icosaxes (harmonic silence), Fibonacci multiplets (SU(4)), the golden interval (tuning) and the Fibonacci stacking word; layer words govern stackings (#92), the Admiral network ground states and the J27/CO tilings. The golden solids in 3D (Icosaxes, Unborn Axes) and in 4D (DICTO All-Golden Polytopes) are all limits of Fibonacci ladders of particle multiplets (SU(4), SO(9)).

6. Methods and Code Map

> *DICTO's idea: a map of which methods the studies share, so that when a method improves, or a fault is found in it, every result that rests on it can be found and re-checked, and so that shared methods can live in one library.*

Scanned: 1324 study scripts. Only 26 import the shared library studies/_lib; most methods exist as copies.

Method	Scripts	Studies	Studies using it (first 8)
harmonic fingerprint (Q_l, W_l)	10	8	2026-10-25-stackings, 2026-11-01-star-cores-book, 2026-11-02-dicto-vacuus, 2026-11-02-vacuus-origami, 2026-11-04-co-rd-sweep, 2026-11-04-harmonic-4d, 2026-11-04-harmonic-fingerprint, 2026-11-04-network-fingerprint
forced-sector search (normal / reversed pieces)	10	8	2026-10-12-bodhi-pieces, 2026-10-16-ocd-pyrochlore, 2026-10-20-double-octet, 2026-10-21-gap-pieces-fill, 2026-11-03-dual-pairs-dimensions, 2026-11-04-pyramid-tilings, 2026-11-04-reversed-neighbour, 2026-11-04-rt-pyramids
separating-axis overlap test	39	16	2026-10-11-self-building-kit, 2026-10-12-dicto-icosa-star, 2026-10-12-network-versions, 2026-10-14-fuji-family, 2026-10-14-tetrapods, 2026-10-15-icosa-star-hollow, 2026-10-16-ocd-pyrochlore, 2026-10-19-og13, 2026-10-21-gap-pieces-fill, 2026-10-22-swept-dodecahedra, 2026-10-25-keyed-web, 2026-10-30-core-cycle, 2026-10-30-spiked-ti, 2026-10-30-star-cores, 2026-11-04-reversed-neighbour, 2026-11-04-rt-pyramids
polar sweep sections S(mu) = P ∩ mu P*	40	21	2026-10-15-icosa-star-hollow, 2026-10-16-interlock, 2026-10-16-ocd-pyrochlore, 2026-10-18-slab-cover, 2026-10-25-atom-onions, 2026-10-25-relay-census, 2026-10-25-stackings, 2026-10-26-janus-kit, 2026-10-27-pyrochlore-halves, 2026-10-29-endless-stack-cell, 2026-10-29-hexoct-cover, 2026-10-29-hexoct-k-clock, 2026-10-29-keyed-hexoct, 2026-10-31-hexoct-ring-build, 2026-11-03-4d-sections, 2026-11-03-6d-golden-cubes, 2026-11-03-dual-pairs-dimensions, 2026-11-03-golden-cuts-beyond-plato, 2026-11-03-knitted-hexoct-patterns, 2026-11-04-co-rd-sweep, 2026-11-04-harmonic-fingerprint
permutations / symmetric point sets	38	27	2026-10-12-bodhi-pieces, 2026-10-12-network-versions, 2026-10-14-cycles, 2026-10-14-fuji-family, 2026-10-16-ocd-pyrochlore, 2026-10-16-polar-dharma, 2026-10-17-scutoids, 2026-10-20-og-spires, 2026-10-20-ott-rays, 2026-10-21-nesting-proof, 2026-10-21-reuleaux-platonic, 2026-10-22-dicto-square, 2026-10-23-shadow-forms, 2026-10-27-five-pyramids, 2026-10-29-golden-cover, 2026-10-30-face-to-face, 2026-10-30-star-cores-exact, 2026-10-30-ti-pyramids, 2026-10-31-star-cores-cover, 2026-11-02-dicto-vacuus, 2026-11-02-vacuus-origami, 2026-11-03-dictonary-cover, 2026-11-04-cubic-pairings, 2026-11-04-harmonic-4d, 2026-11-04-harmonic-fingerprint, 2026-11-04-network-fingerprint, _lib
exact checks (PASS/FAIL ledger)	64	36	2026-10-14-axe-cleo-fuji, 2026-10-19-five-fold-octet, 2026-10-19-kepler-poinsot-up, 2026-10-21-nesting-proof, 2026-10-22-platonic-mix-step1, 2026-10-22-swept-dodecahedra, 2026-10-23-shadow-forms, 2026-10-25-atom-onions, 2026-10-25-bond-physics, 2026-10-25-changeover-evidence, 2026-10-25-gnomon-shapes, 2026-10-25-physical-anchors, 2026-10-25-relay-census, 2026-10-25-scale-anomalies, 2026-10-25-stackings, 2026-10-25-web-of-bonds, 2026-10-26-janus-kit, 2026-10-27-splash-rules, 2026-10-27-white-queen-600, 2026-10-27-white-queen-polar, 2026-10-29-endless-stack-cell, 2026-10-29-hexoct-cover, 2026-10-29-hexoct-k-clock, 2026-10-29-keyed-hexoct, 2026-10-30-star-cores-exact, 2026-11-02-hexoct-connected, 2026-11-02-hexoct-knit-mechanics, 2026-11-03-6d-golden-cubes, 2026-11-03-dual-pairs-dimensions, 2026-11-03-golden-cuts-beyond-plato, 2026-11-03-hexoct-cushion-plan, 2026-11-04-cross-links, 2026-11-04-cubic-pairings, 2026-11-04-cuboctahedron-pyramids, 2026-11-04-pyramid-tilings, 2026-11-04-trd
rotations about an axis	57	34	2026-10-11-self-building-kit, 2026-10-11-spheres-and-ripples, 2026-10-12-self-building-ring, 2026-10-14-fuji-family, 2026-10-15-icosa-star-hollow, 2026-10-16-ocd-pyrochlore, 2026-10-20-double-octet, 2026-10-21-aristotle-pyramids, 2026-10-21-og-turned-out, 2026-10-22-dicto-square, 2026-10-22-platonic-mix-step1, 2026-10-22-swept-dodecahedra, 2026-10-23-exact-vs-dissection, 2026-10-23-shadow-forms, 2026-10-24-keyed-w-penrose, 2026-10-25-keyed-web, 2026-10-29-golden-cover, 2026-10-29-quanta-atoms, 2026-10-29-wonder-cover, 2026-10-30-core-cycle, 2026-10-30-star-cores, 2026-10-30-star-cores-exact, 2026-10-31-star-cores-cover, 2026-11-02-dicto-vacuus, 2026-11-02-hexoct-knit-mechanics, 2026-11-02-vacuus-origami, 2026-11-03-6d-golden-cubes, 2026-11-03-bonds-cover, 2026-11-03-dictonary-cover, 2026-11-03-dual-pairs-dimensions, 2026-11-03-golden-cuts-beyond-plato, 2026-11-04-reversed-neighbour, 2026-11-04-rt-pyramids, 2026-11-04-trd

Functions defined separately in three or more studies (most copied first): faces (78); check (64); rot (53); run (40); build (40); sheet (39); describe (35); planes (34); draw (29); overlap (28); solid (27); main (27); edges (26); section (25); cyc (24).

Plan: move the core methods (harmonic fingerprint, forced-sector search, overlap test, polar sweeps, symmetric point sets, exact-check ledger) into studies/_lib with their own tests and calibrations; new studies import them; finished studies keep their copies so their published results stay reproducible. Matching is by function name and approximate (generated by code_map.py).

6.1 Tests against families: sets, subsets and gaps

— = not yet run (a gap in understanding, or outside the test's competence). DICTO 4 Nov 2026: tests and methods group the results into sets and subsets and show where further gaps lie. · 4 Nov: cells outside a test’s competence marked n/a (DICTO). · 4 Nov: column "Halves (ortho / gyro)" added (DICTO: half pieces were a blind spot); classical operation (Johnson 1966 ortho/gyro; gyrated honeycombs).

Family	Harmonic Q4/Q8 (hidden symmetry)	Harmonic Q6	Isoperimetric (roundness)	Mahler (polar balance)	Foam wall area	Forced-sector search (tiling)	Dehn (space-filling screen)	Interlocking (kinematic)	Halves (ortho / gyro)
Cube–octahedron sweep	Icosaxes φ, exact (Q4, Q8); bronze (Q8)	zeros not at stations	√3 exact = Lindelöf (all facets tangent to one sphere); √n in n dimensions	max near φ, cubic; ends 32/3	n/a (outside the test's competence)	n/a (outside the test's competence)	zero only at Kelvin 3/2 and the cube (the lone tilers)	n/a (outside the test's competence)	twist convex only at the CO (→ J27); TO twist dented
CO–RD sweep	one Q4 zero, not golden	one Q6 zero, μ ≈ 1.65 (not golden)	max ≈ 1.523, no station	max $12\mu^3-36\mu^2+43=0$, no station	n/a (outside the test's competence)	only Dehn-zero member (B3 permutohedron) killed by angle count	zero for $1 \le \mu \le 4/3$ (cube-trimmed RDs, rational angles) and at μ = 5/3 (B3 permutohedron); none tile (angle count) — corrected 4 Nov	n/a (outside the test's competence)	twist convex along the whole sweep (RD→TRD … CO→J27); B3 permutohedron twist convex, Dehn nonzero
Tetrahedron sweep T ∩ μ(−T)	Q4 zero at a quartic root, not golden	never silent (no zero on the sweep; Q4 zero at μ ≈ 1.8055 and its mirror)	max exactly at the octahedron (Lindelöf: facets tangent to one sphere)	$f = 4(\mu^3-9\mu^2+27\mu-11)/(3\mu)$; octahedron a local min 32/3; max at $(11-\sqrt{33})/4$, no station	n/a (outside the test's competence)	μ = 2 tiles alone (octet exact cover, N = 20, two orientations) — KNOWN: Damasceno–Engel–Glotzer 2012 (β-tin)	zero only at μ = 2 (and 1/2): T(4) minus corners	n/a: the β-tin tiling has no flat one-tile-thick layer (jagged slabs gave 20/31 locked, free tiles = protruding ones)	—
Delian walk (#51)	Q4 at t = 1/9 exactly; no joint Q4/Q8	T piece Q6 zeros	max at $t = 1-\sqrt{6}/3 \approx 0.1835$ = Lindelöf point (all faces tangent), checked to grid 0.0005	max ≈ 13.852 at t ≈ 0.135, no station	n/a (outside the test's competence)	rigid except octet (#93)	O(t): nonzero for all t > 0 (one irrational angle 144.74°, no cancellation); zero only at the RD	n/a (outside the test's competence)	—
CsCl pairs (#90)	as the sweep: Icosaxes φ silent (pieces are cube–octahedron sweep members P_μ, P_{3−μ}: see that row)	as the sweep (pieces are cube–octahedron sweep members P_μ, P_{3−μ}: see that row)	each piece as the sweep (√3 = Lindelöf); Kelvin pair both at 3/2 (pieces are cube–octahedron sweep members P_μ, P_{3−μ}: see that row)	each piece as the sweep (pieces are cube–octahedron sweep members P_μ, P_{3−μ}: see that row)	max at Kelvin 3/2, min at CO+O	rigid (#93 Thm A)	pair sum zero (they tile); a single piece zero only at Kelvin μ = 3/2 (the equal pair)	n/a: no flat one-tile-thick layer (same reason)	—
4D sweep tesseract ∩ μ·16-cell	Q4 at 24-cell (μ = 2)	Q6 at √2, 1+√3 exact	max exactly at the 24-cell μ = 2 = √n (Lindelöf); all n: √n	max ≈ 16.747 at μ ≈ 2.2837, no station; 24-cell 16; ends 32/3	n/a (outside the test's competence)	n/a (outside the test's competence)	n/a (3D screen; 4D Dehn invariants not built)	n/a (outside the test's competence)	—
600-cell slices	corner-first always silent; cell-first never jointly	corner-first silent at 3 heights; path between the Queens: Q12 dip ≈ 0.925, Q20 dip ≈ 0.887	corner-first max 0.9747 at h ≈ 0.163; cell-first 0.9708 at h ≈ 0.104; no station	corner-first max 17.077 at h ≈ 0.170; cell-first 16.979 at h ≈ 0.090	n/a (outside the test's competence)	n/a (outside the test's competence)	nonzero at all 800 heights tested	n/a (outside the test's competence)	—

Family	Harmonic Q4/Q8 (hidden symmetry)	Harmonic Q6	Isoperimetric (roundness)	Mahler (polar balance)	Foam wall area	Forced-sector search (tiling)	Dehn (space-filling screen)	Interlocking (kinematic)	Halves (ortho / gyro)
		(preliminary)							
Centre pyramids T, C6, R12, RT	none silent (T 0.528, C6 0.721, R12 0.632)	none silent (0.52, 0.30, 0.25)	T 0.110, C6 0.223, R12 0.329 (RT pyramid not computed)	T 64/9 (simplex); C6 = R12 ≈ 8.681 (affinely equal square pyramids)	n/a (outside the test's competence)	T, C6 whole solids; R12 free + slides; RT open	zero for C6, R12, RT (cube, RD, RT the only isohedral)	base-outward on a free surface: free; otherwise locked	—
Admiral Cap network	nodes: fcc vs twin pattern heard	A cap 0.358, I 0.663, Q′ 0.524: none silent	A cap 0.414 (I 0.829, Q′ 0.110)	A cap 9.000168 (not 9); I 14.076; Q′ 64/9	n/a (outside the test's competence)	frustration 8/12; ground-state layers	all three kinds nonzero alone (Q′, I, A cap): only the mix can fill	slab: all caps and icosahedra locked (slides and twists)	—
Golden nests / OG / AXE networks	OG family: Q4 = Q8 = 0 (icosahedral); W6 sign splits corner-type (TQHD, TQP, OG, KRE) from face-type (TQH, TPH, PHE, TPP, DPHP), changing in pairs	OG family lowest Q6: TQP 0.032, DPHP 0.037, PHE 0.039, KRE 0.041 (none silent)	OG family IQ 0.938 (OG) … 0.969 (DPHP, roundest)	16.59 (OG) … 17.13 (DPHP); ball 17.55	n/a (outside the test's competence)	none: Dehn nonzero for all nine OG solids	OG family (9 solids): all nonzero — none tiles alone (calibrated screen, 11/11)	n/a (outside the test's competence)	—
HexOct tiles (#75)	no Q4 silence on the λ walk (min 0.025 h, 0.167 c)	no silence; jump at 5/24 from the corner-set change (artefact; same event as the Dehn onset)	fixed h-piece (λ ≥ 3/8): 0.487	fixed h-piece: exactly 34/3	n/a (outside the test's competence)	n/a (outside the test's competence)	EXACT: h pieces $36(5/24-\lambda)\otimes\beta$ (zero from 5/24 on); c pieces $24(3/8-\lambda)\otimes\beta$ (zero at 3/8); tiles zero	one layer of c-tiles or h-tiles: all locked (slides and twists)	—